

IS THERE A PLATONIC TREE?

CALIBRATED CLASS GEOMETRY IN FOUNDATION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Foundation models are often said to converge toward shared representations. We ask this question of geometry: is a model’s arrangement of classes tree-like, and is that tree shared? Raw tree-likeness (Gromov δ) is confounded by dimension, class count and covariance, so we build a calibrated instrument that reads every score as an excess over a random cloud with the same dimension and spectrum. Across 12 vision backbones, 6 datasets and 16 text models the instrument gives a nuanced answer. Structure beyond the random baseline is nearly universal in vision (68 of 72 cells) and, in text, depends on the training recipe and emerges with scale; a star calibration shows that this structure is clustered rather than deep: models organize classes around hubs, with little hierarchy beyond, whether or not they were trained in hyperbolic space. The trees are moderately shared: a naive comparison of the models’ hierarchies isolates the self-supervised family as an island, an artifact of the clustering step; once controlled, every training recipe recovers the human taxonomy partially, more so with supervision. The same calibrated scores also indicate, with modest gains, where non-Euclidean readouts help. What foundation models share is clustered structure and a partial, supervision-dependent recovery of the human taxonomy — not one common tree.

1 INTRODUCTION

Is there a Platonic tree? Foundation models organize classes into calibrated, clustered structure and partially recover the human taxonomy, more so with supervision. They do not converge to one common tree.

figures/fig1_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

Figure 1: Raw tree-likeness cannot tell a random cloud, a star and a tree apart; the calibrated instrument can. A high-dimensional random cloud, clusters around a hub, and a nested hierarchy all give a low raw δ . The excess over a spectrum-matched null separates the random cloud from the other two; only the matched-star test separates the star from the tree.

Semantic knowledge is hierarchical: instances belong to categories (a husky is a dog), categories to superordinate classes (dogs are mammals), and so on upward. The natural geometric home for such hierarchies is hyperbolic space (Nickel & Kiela, 2017; Sarkar, 2011), and a long line of work

imposes it at training time (Khrulkov et al., 2020; Chen et al., 2022; Desai et al., 2023). A complementary observation, latent hyperbolicity, holds that standard Euclidean models already organize concepts in a tree-like metric structure (Khrulkov et al., 2020; Yang et al., 2026). Is that structure hierarchical, and for whom?

The naïve measurement of this question is confounded. Gromov δ -hyperbolicity (Gromov, 1987) assigns 0 to trees, but three artifacts push raw readings down without any hierarchy: high-dimensional clouds concentrate toward equidistance (a degenerate star tree), anisotropic spectra mimic low-dimensional behavior, and small class counts shrink the quadruple population. These artifacts are real: small GPT-2 models sit *at* their matched null despite low raw δ , and classic sentence embedders share raw δ with vision-contrastive models while carrying none of their genuine structure.

Our first contribution is an **instrument**: the exact estimator calibrated on reference geometries (trees, hyperbolic regions, spheres, Gaussians of growing dimension), with every reading expressed as an *excess* over a null cloud matched in dimension and covariance spectrum and ranked against 200 null replicates. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared — is there a Platonic tree?

What survives is clustered form. Structure beyond the matched null is nearly universal in vision (68 of 72 model $\times$ dataset cells) and emerges with recipe and scale in text. A star calibration shows what that structure is: classes organized around hubs, with depth beyond a matched star at most marginal, whether or not the model was trained in hyperbolic space.

Sharing is graded, and its naive measurement misleads. A naive tree-to-tree comparison isolates the label-free family as a categorical island, an artifact of the clustering step: once the configuration is controlled, ImageNet agreement rises from 0.03 to 0.38. Every training recipe recovers the human taxonomy partially, more so with supervision, and no configuration returns one common tree.

Our contributions are:

1. **A calibrated instrument for tree-likeness.** Estimator calibration on reference geometries, spectrum-matched nulls read as percentile ranks, and a class-count control. Identical raw readings receive opposite verdicts, so the raw scale is not interpretable.
2. **A calibrated census.** Genuine tree-like form in nearly every vision cell and, in text, form that depends on recipe and emerges with scale. A star calibration shows that the excess certifies clustering, not depth.
3. **The map of trees.** The naive comparison manufactures a categorical island. The controlled comparison shows graded, supervision-dependent recovery of the human taxonomy.
4. **A scoped corollary:** the same calibrated descriptors indicate where non-Euclidean readouts pay on prototype-based tasks.

2 RELATED WORK AND BACKGROUND

Hyperbolic deep learning: imposing vs measuring. Building on tree embeddability (Sarkar, 2011) and Poincaré embeddings (Nickel & Kiela, 2017; Ganea et al., 2018), hyperbolic methods span vision (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024), vision-language (Desai et al., 2023; Pal et al., 2025), graphs (Chami et al., 2019), and language (Chen et al., 2022; He et al., 2026; Sinha et al., 2024). Latent hyperbolicity, low measured δ in standard models, has been reported for CNN/ViT features (Khrulkov et al., 2020; Bdeir et al., 2024), token embeddings (Yang et al., 2026), and word vectors (Tifrea et al., 2019). Those readings are sample-level and uncalibrated; transplanted to sample-level features, our instrument finds excesses mostly within null noise (Appendix A.16), and our census is about inter-class (centroid) geometry.

Geometric analyses of representations. Representation similarity (Kornblith et al., 2019), intrinsic dimension (Ansuini et al., 2019; Pope et al., 2021), and neural collapse (Papayan et al., 2020) characterize Euclidean structure without testing tree-likeness. Estimating Gromov hyperbolicity on sampled quadruples has known bias and scale caveats (Fournier et al., 2015), which our calibration-plus-null design addresses operationally.

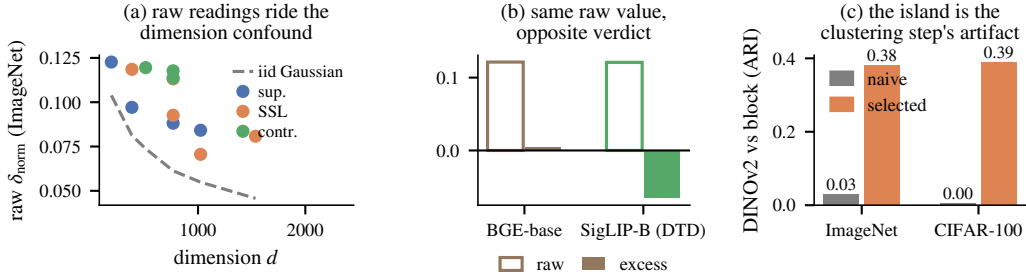


Figure 2: **Why calibration: raw δ cannot be read on its own, and naive tree comparison manufactures islands.** (a) Raw δ_{norm} rides the dimension confound: iid Gaussians (dashed) drift low as d grows, and the 12 backbones’ ImageNet readings sit on the same slope. (b) A text embedder (BGE-base) and a vision cell (SigLIP-B on DTD) share the same raw value with opposite null verdicts. (c) DINOv2-vs-block tree agreement before and after the configuration control of §5.

Representational convergence. The Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations; Koepke et al. (2026) stress-test cross-modal convergence at scale, and Gröger et al. (2026) show that permutation-calibrated similarity removes most apparent global convergence while local neighborhood structure survives. Park et al. (2024) read hierarchical concept structure in LMs from the orthogonality of concept directions; our text census is a complementary, metric-level reading of the same question. We are the geometric complement of this program: the same calibration philosophy applied within models and, in §5, across them.

Background. For four points w, x, y, z of a metric space, sort the three pairing sums $d(w, x) + d(y, z)$, $d(w, y) + d(x, z)$, $d(w, z) + d(x, y)$ as $S_{\text{max}} \geq S_{\text{mid}} \geq S_{\text{min}}$; the four-point defect is $(S_{\text{max}} - S_{\text{mid}})/2$ and δ is its supremum. Trees have $\delta = 0$; under this four-point definition the hyperbolic plane has $\delta = \ln 2 \approx 0.693$ (Nica & Špakula, 2016), distinct from the thin-triangles constant $\ln(1 + \sqrt{2}) \approx 0.881$. We use the normalized $\delta_{\text{norm}} = \delta_{\text{max}}/\text{diam}$ throughout: on this scale hyperbolic regions drift to 0 as they grow while spheres stay high, so the intuitive ordering trees $<$ hyperbolic $<$ spherical holds. Ollivier–Ricci curvature (Ollivier, 2009) on a kNN graph is our local companion: positive on tightly clustered neighborhoods, negative on tree-like bridges.

3 THE INSTRUMENT

This section builds the measurement layer in whose units every later claim is stated: raw δ_{norm} is interpretable only against calibrated references and matched nulls.

Classes are centroids; a cell is a (model, class set) pair. Each frozen model is a fixed feature extractor; classes are represented by centroids (vision: the mean of each class’s cached training embeddings, 100 on ImageNet, the full cached split of 80–6,000 elsewhere; text: embeddings of “a photo of a {class}” over the 1000 ImageNet classes, plus the corpus variations of §4). The calibrated census covers 12 vision backbones $\times$ 6 datasets (a *cell* is one model $\times$ dataset pair) and 16 text models (panel in Appendix A); the task grid covers the 10 vision backbones with cached per-dataset features (all but DINO-B and SigLIP-B). Estimator settings: the supremum four-point statistic is the pre-registered primary reading and the 99.9th percentile is reported only as robustness; 5×10^5 quadruples per seed, 10 seeds (exact enumeration when $C \leq 30$); ORC on $k=10$ kNN graphs (ranking invariant for $k \in \{5, \dots, 50\}$). On the transfer sets the excess is budget-stable from 10^5 quadruples and image-resampling moves it far less than its size; on ImageNet every sign holds at every budget but magnitudes drift by up to 0.02; a 99.9th-percentile variant preserves every sign (Appendix A.13).

A low raw value is not evidence. On reference geometries (Table 3, Appendix A.2) the estimator recovers the tree zero and the $\ln 2$ four-point constant of $\mathbb{H}^2$ ($\delta_{\text{norm}} \rightarrow 0$ as hyperbolic regions grow),

spheres stay high (0.14–0.18), and iid Gaussians drift low as d grows (0.104 $\rightarrow$ 0.046 from $d=192$ to 1536): the dimension confound. A low raw value is not evidence.

The reading is an excess over a matched null. For every (model, class set) cell we synthesize null clouds with the same n , d , mean and covariance *spectrum* (Gaussian coefficients recombined with the real singular values; alternative constructions in Appendix A.13), and report the **excess** $\delta_{\text{norm}}^{\text{real}} - \delta_{\text{norm}}^{\text{null}}$ over 200 spectrum-null replicates (20 for ξ). The primary evidence is the *percentile rank* $r/200$, the number of replicates above the real value, and its left-tail $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; a cell is *genuine* when $p \leq 0.05$, i.e. $r \geq 191$. Rank-based calibration is what gives permutation scores their finite-sample validity (Gröger et al., 2026, §5.1); a z -score on a handful of replicates is not rank-based, so z is kept only as a secondary summary in the appendix. The null is aggregation-aware in the sense of Gröger et al. (2026, §5.2): the same supremum over the same quadruple budget (5×10^5 per seed) is evaluated on every null replicate, so the max-statistic inflation is calibrated rather than ignored; the real cloud averages 10 quadruple seeds and each replicate 5 (3 in the text census), which affects variance, not bias. “Tree-like” below always means *negative excess*, and “genuine” means $p \leq 0.05$ against the spectrum-matched Gaussian null, one hypothesis of “no hierarchy” (each null’s scope: Appendix A.1). Sign-positive excess means *less* tree-like than chance given the spectrum.

ξ reads norm structure, not curvature. A parallelogram estimator ξ (midpoint defect of triangles) was intended as a curvature sign, but references with heterogeneous norms and no curvature drive it negative (Appendix A.19); we therefore report it only as a descriptor of norm structure, never as curvature.

Class count is a measured variable, not a nuisance. Because C sets the quadruple population, we sweep ImageNet subsets $C \in \{10, \dots, 1000\}$ with two compositions: random (spanning the full hierarchy) and WordNet-coherent (siblings, effectively *flat*). Results in §4; the control makes “needs a real hierarchy” a measured variable.

One parameter-free projection and two tasks serve the corollary. The parameter-free map $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$ centers features, scales the 95th-percentile norm to radius $1/\sqrt{2}$, and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

4 FINDINGS I: STRUCTURE BEYOND THE NULL

Vision: structure beyond the null in 68 of 72 cells. Table 1 summarizes the census. Sign: 68 of 72 cells have negative excess. The sign-positive cells are DINOv2-S/B/G on ImageNet, above their null by a small margin (excess $\leq +0.011$, $p \geq 1.00$) yet genuine under the robustness statistic (Appendix A.13), and ViT-T on FashionMNIST, at zero excess. Genuine cells ($p \leq 0.05$): 55 of 72 overall and 43 of 48 on the four hierarchical-label datasets (ImageNet, CIFAR-100, CIFAR-10, DTD); 44 cells lie below all 200 replicates. The flat datasets are sign-consistent (23/24) but sit near the estimator’s resolution limit at $C=10$, so only 12 of 24 are genuine (class-count control below; per-cell ranks and p in Appendix A.4). The magnitude, not the sign, of the small- C excesses depends on how the null is built: an exact-sample-spectrum construction shrinks DINOv2’s CIFAR-100 excess while signs agree in 35/36 cells (Appendix A.13). Two ResNet-50 self-supervised models from the centroid store (Barlow Twins, BYOL) also sit below their null on ImageNet (Table 7), so the form is not a ViT artifact. On ImageNet, supervised ViTs sit at excess -0.020 to -0.030 and CLIP/SigLIP at -0.016 to -0.025 ; DINOv2-L is genuine on ImageNet with a small excess (-0.008 , $r = 200/200$), while within each transfer dataset the same family’s excess deepens with scale (S $\rightarrow$ G on CIFAR-10: $-0.095 \rightarrow -0.155$; per-dataset trends in Figure 3b). Because excess magnitudes shrink with C (class-count control below), cross-dataset comparisons are not meaningful; all statements here are within-dataset trends. Under the supremum-robust p99.9 statistic the census itself is unchanged, 70/72 sign-negative and 42/48 hierarchical cells genuine at 20 replicates,

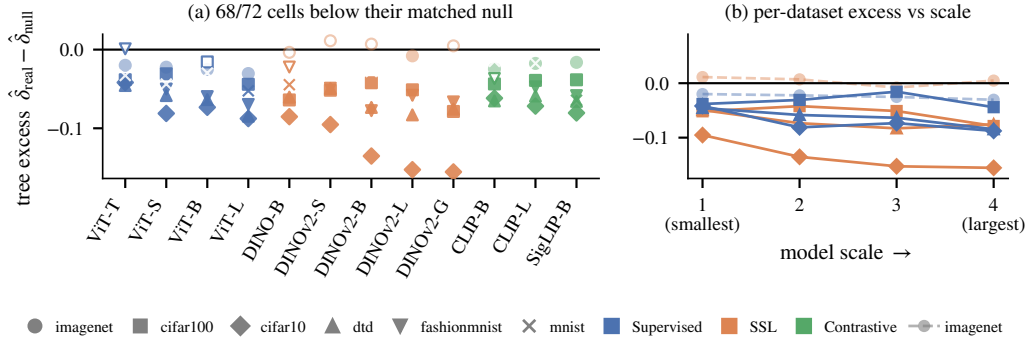


Figure 3: **Tree-like structure beyond the matched null is nearly universal in vision.** (a) Excess per vision cell (12 models $\times$ 6 datasets; hollow markers: not genuine, $p > 0.05$); the exceptions are DINOv2-S/B/G on ImageNet, above their null by less than the estimator noise. (b) Per-dataset excess vs scale: transfer-set excesses (solid) deepen with scale within each dataset, DINOv2’s ImageNet supremum excess (dashed) stays at null, and cross-dataset magnitudes are not comparable.

model	family	δ_{norm} IN	excess IN	$r/200$ (p)	exc. C100	exc. C10	ρ_{WN}
ViT-T	sup.	0.123	-0.020	199 (.010)	-0.038	-0.042	+0.52
ViT-S	sup.	0.097	-0.022	200 (.005)	-0.031	-0.081	+0.50
ViT-B	sup.	0.088	-0.025	200 (.005)	-0.016°	-0.073	+0.49
ViT-L	sup.	0.084	-0.030	200 (.005)	-0.044	-0.088	+0.53
DINO-B	SSL	0.113	-0.003	162 (.194)	-0.064	-0.085	+0.36
DINOv2-S	SSL	0.119	+0.011	0 (1)	-0.052	-0.095	+0.22
DINOv2-B	SSL	0.093	+0.007	0 (1)	-0.042	-0.135	+0.18
DINOv2-L	SSL	0.071	-0.008	200 (.005)	-0.051	-0.153	+0.18
DINOv2-G	SSL	0.081	+0.005	1 (.995)	-0.079	-0.155	+0.20
CLIP-B	contr.	0.120	-0.025	200 (.005)	-0.044	-0.062	+0.57
CLIP-L	contr.	0.118	-0.018	200 (.005)	-0.039	-0.072	+0.59
SigLIP-B	contr.	0.114	-0.016	199 (.010)	-0.038	-0.080	+0.58

Table 1: **The calibrated census: genuine tree-like structure in nearly every vision cell.** One row per backbone (supervised / self-supervised / contrastive blocks). δ_{norm} : raw ImageNet reading; excess: δ_{norm} minus the spectrum-matched null mean; $r/200$: number of the 200 null replicates above the real value, with the left-tail $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; a cell is *genuine* when $p \leq 0.05$ ($r \geq 191$); bold: sign-positive, i.e. less tree-like than the null; ° on a transfer-set excess: not genuine ($p > 0.05$). ρ_{WN} : Spearman correlation of inter-centroid and WordNet distances. Cross-dataset magnitudes are not comparable (§4). DTD, the flat datasets, the p99.9 robustness statistic and the task columns are in Appendix A (Table 5).

but the ImageNet ordering shifts: DINO/DINOv2 become genuine while ViT-T’s supremum excess does not survive (Appendix A.13). Under the primary statistic DINOv2 on ImageNet therefore sits at its null; under the robustness statistic it is genuine. We report both and do not adjudicate; whose tree it is (§5) does not depend on the choice.

The excess certifies clustering, not depth. The spectrum excess measures structure beyond second moments, and a pure star already has such structure: a synthetic 30-cluster star with no hierarchy scores ≈ -0.11 , as deep as any real cell (Appendix A.5). Depth is therefore tested separately, by comparing each backbone with a *matched star* that copies its hub radius, its within-superclear spread and its cluster sizes (Table 9). The matched-star test finds residual depth marginal: on ImageNet only DINOv2-L beats its star beyond the noise, and on CIFAR-100 every backbone is *less* tree-like than its star. A backbone *trained* in hyperbolic space is no exception: MERU (Desai et al., 2023) and its Euclidean CLIP twin show the same clustering excess and the same absence of residual depth, and MERU’s Lorentz-metric $\hat{\delta}$ matches its Euclidean value (Appendix A.5). The excess measures how

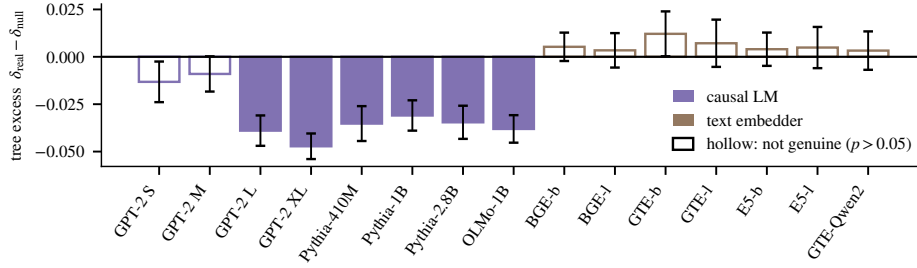


Figure 4: **In text, genuine form depends on recipe and emerges with scale.** Calibrated text census (padding-free extraction, 200 null replicates; error bars combine null and estimator s.d.; hollow bars are not genuine, $p > 0.05$; OLMo-7B, not re-extracted, omitted). Form emerges with scale in GPT-2, is constant in Pythia, and is absent in classic sentence embedders despite raw δ equal to genuinely structured vision models.

classes are arranged, not the geometry they are embedded in; the class-subsampling control below measures the hierarchy’s reach, and *whose* tree it is the subject of §5.

Text: the instrument reveals what raw δ hides. Within causal LMs, genuine form *emerges with scale* in GPT-2 (Figure 4): under padding-free extraction S and M are not genuine ($-0.013/-0.009$) while L and XL are ($-0.039/-0.047$; Table 12). Pythia carries genuine form at every scale tested, and so does OLMo-1B. Extraction is part of the measurement: left-padded batching alters GPT-2’s hidden states and moves $\hat{\delta}$ by up to 0.02, of the order of the marginal excesses (Table 13); the padding-free protocol removes this artifact, the scale trend survives it, and individual marginal cells (GPT-2 M) do not. The rising nulls track the effective rank of the centroid covariance, and genuineness is not a proxy for anisotropy (GTE-base has the highest rank and sits at its null); a PC-permutation null confirms every verdict. Classic sentence embedders (BGE/GTE/E5) are not genuine despite raw values indistinguishable from vision-contrastive models whose excess is: identical raw values, opposite verdicts (Figure 2b). The same embedders show genuine excess (-0.020 to -0.028) on DBpedia, a real three-level text hierarchy: presence of form is a property of the (model, concept set) pair. The only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine either, and we do not build on it (Table 12). Causal-LM values are template-sensitive but the scale ordering is template-robust; per-template values, ranges and the padding-free re-run are in Appendix A.15.

Hierarchy depth, not class count. At fixed C , random ImageNet subsets, which span the full hierarchy, are consistently more tree-like and more exploitable than WordNet-coherent sibling subsets, an effectively flat concept set (DINOv2-L at $C=50$: excess -0.124 vs -0.093 ; Appendix A). Raw δ_{norm} rises with C , so MNIST’s flat verdict is not a class-count artifact. DINOv2-G reaches its null at $C \geq 500$ on ImageNet, reproducing the census exception.

The form tracks training. Two interventions move it (Figure 5; raw δ , within architecture): fine-tuning on a task with no class hierarchy raises δ by 19–91%, and δ falls across transformer depth. Shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should. Random-weight networks give higher raw δ , but their features are near-Gaussian, so this is expected by construction and not counted as evidence; excess-based versions of these interventions need the stored point clouds and are left to future work. On matched architecture (ViT-B supervised/DINO/CLIP) all three objectives yield genuine form on most datasets, and the family separations emerge with scale rather than at fixed size.

Negative curvature concentrates on bridges. High mean ORC coexists with tree-likeness because they live on different edges: negative curvature concentrates on between-superclass bridges (about twice as often as within superclasses on ImageNet; Appendix A), while the mean measures within-cluster compactness. DINOv2 on ImageNet inverts the pattern, consistent with its at-null supremum reading, and recovers it on CIFAR-100.

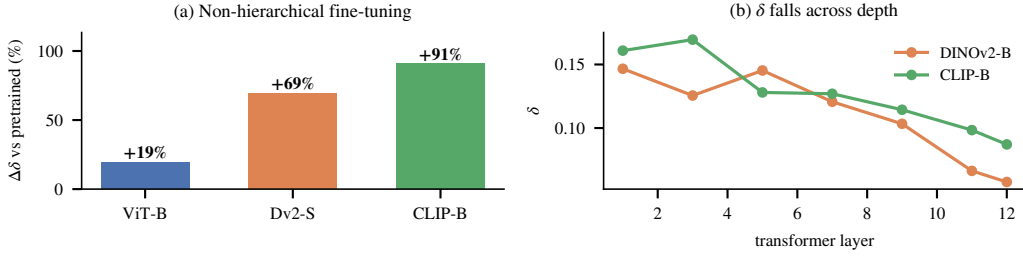


Figure 5: **The form is learned: training interventions move it.** Raw δ , within architecture. (a) Non-hierarchical fine-tuning raises δ 19–91%. (b) δ falls across transformer depth (layers ≥ 1).

5 FINDINGS II: SHARED CONTENT, FRAGILE MEASUREMENT

If the measured trees were a shared representation of reality, models should agree with each other and with human taxonomy. Whether they appear to depends on how the trees are measured: the naive-vs-calibrated gap is the paper’s thesis in miniature.

The naive map manufactures an island. We extract each model’s hierarchy explicitly (a dendrogram over the shared centroids) and compare hierarchies, not representations, by ARI between cut partitions (adjusted Rand index, chance-corrected partition agreement) and by the cophenetic correlation (CPCC) between tree and distances. Under the naive configuration, Euclidean distances with average linkage, the map (Figure 6) is spectacular: supervised ViTs and contrastive VLMs form one block (cross-family ARI up to 0.82), while the label-free DINOv2 family appears as an island (ARI ≤ 0.06 against every supervised model). The categorical island, however, is a measurement artifact. Its cuts are degenerate: DINOv2’s Euclidean average-linkage dendrograms chain (up to 97% of classes in one cluster), forcing ARI ≈ 0 by construction, and Euclidean distance is the metric our diagnostics flag as mismatched for this family (§4). A configuration must be selected by a criterion fixed *without reference to the agreements being compared*. Cophenetic fidelity alone re-selects the chaining configuration, so we fix: *highest CPCC among configurations whose reference cuts are non-degenerate for every model* (largest cluster $\leq 50\%$); it selects cosine-average on ImageNet and cosine-complete on CIFAR-100 (Appendix A.8). The 0.5 degeneracy bound is the criterion’s one free parameter: across admissible bounds the ImageNet gap ranges from 0.13 to 0.38, never back to the categorical island (≤ 0.06 ; Appendix A.8). Under the selected configurations (Figures 6b and 7b) the island shrinks from categorical to moderate (ImageNet 0.38 vs block 0.48). Across *all* non-degenerate configurations DINOv2 recovers the true 20 CIFAR-100 superclasses at ARI 0.2–0.5, reaching the supervised band only under cosine-Ward, the configuration where the CIFAR-100 island also closes fully (per-configuration values in Appendix A.8).

The island is an artifact; sharing is graded. Three statements hold across every non-degenerate configuration. The categorical island is an artifact. Label-free models recover the human taxonomy partially, and supervision deepens the resemblance: paired within each admissible configuration, the label-supervised ViT recovers more than each DINOv2 model in 11 of 12 comparisons (Appendix A.8). Tree-to-tree comparison itself is fragile to configuration. A moderate label-free gap persists on ImageNet under every configuration, consistent with the same reading: a real tree, least aligned with WordNet at the top.

WordNet alignment follows supervision, not tree-likeness. Spearman correlation between inter-centroid distances and WordNet tree distances on ImageNet: +0.57 to +0.59 for CLIP/SigLIP, +0.49 to +0.53 for supervised ViTs, +0.36 for DINO-B, and only +0.18 to +0.22 for DINOv2, the most tree-like family, with shuffle and Gaussian controls at ≈ 0 (Table 1). Cutting centroid dendrograms recovers the 20 true CIFAR-100 superclasses at ARI up to 0.61 (permuted labels ≈ 0). On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 38). Three controls bound circularity: WordNet-coherent groupings are more tree-like than random ones in 11/12

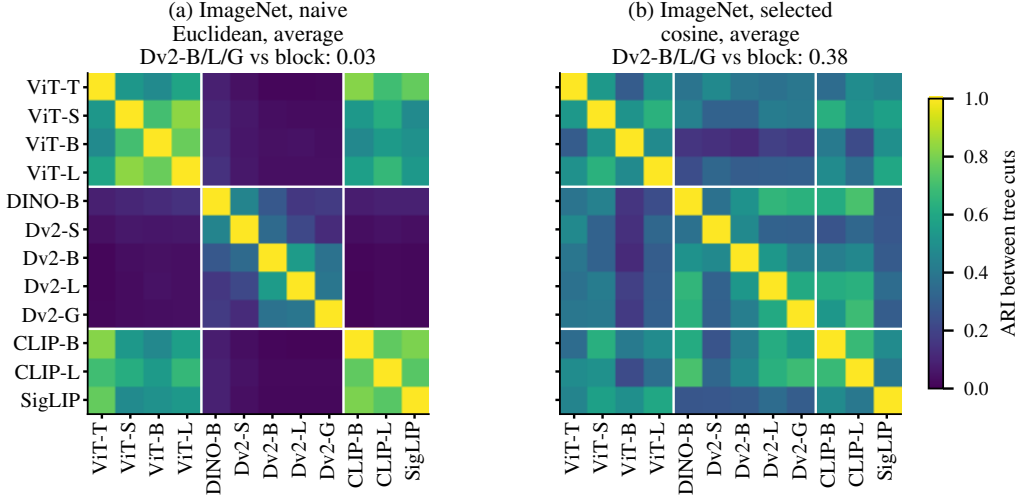


Figure 6: **The categorical island vanishes once the clustering step is controlled.** Pairwise ARI between dendrogram cuts of the 12 vision models on ImageNet (30 clusters): naive configuration (a: Euclidean, average linkage) vs criterion-selected (b: cosine, average), with the mean DINOv2-B/L/G-vs-block ARI above each panel. Per-configuration values in Appendix A.8; the CIFAR-100 pair, same reading, in Figure 7.

models; single-image distances still correlate with WordNet (ρ up to $+0.26$); and the decoupling above is the opposite of label injection. DINOv2’s semantics concentrate in the angular component (sibling-triplet agreement 0.91 under cosine vs 0.71 Euclidean for DINOv2-L; Appendix A), which is also why Euclidean dendrograms chain for this family and produce the naive island. The census exception is consistent with this: whatever structure DINOv2 has on ImageNet (at null under the primary statistic, genuine under the robustness one), it is the least aligned with WordNet’s top level: the form-vs-content split.

The recovery is not WordNet circularity. ImageNet classes are WordNet synsets, so recovering WordNet there could be circular. Repeating the tree map on DBpedia Classes, a three-level ontology independent of WordNet (219 leaf classes; the three text embedders), closes this door: no configuration degenerates, the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 across all six configurations, and they agree with each other at cross-model ARI 0.65–0.82. The recovered non-WordNet hierarchy is real, shared, and configuration-robust; the fragility of §5 is a property of the (family, concept set) pair.

Local agreement survives calibration. We apply the permutation calibration of Gröger et al. (2026) as they define it: for each of the 66 model pairs, $K=200$ permutations of the class correspondence give the null distribution, the threshold $\tau_{0.05}$ is its $[0.95(K+1)]$ -th order statistic, and the calibrated score is $\max\{(s - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$ (scalar calibration, no layer search). Local neighborhood agreement survives: mutual-kNN is 0.472 raw and 0.464 calibrated, with every pair at $p = 1/201$, and the null mean 0.010 equals the analytic chance level $k/(n-1) = 10/999$. Centroid CKA survives too (0.633 raw, 0.577 calibrated, null mean 0.108): centroids distill a genuinely shared relational base. The Poincaré reading improves neither agreement (kNN 0.472 $\rightarrow$ 0.425; CKA 0.633 $\rightarrow$ 0.640): hyperbolic distances buy no cross-model alignment, as expected, since ϕ preserves directions and rescales norms monotonically (per-pair values and the null-centered variant in Table 21).

Form and content anti-align. The families with the most WordNet content (supervised ViTs, CLIP) carry the most fragile form, borne by the supremum statistic. The family with the most robust form (DINOv2) is the least aligned with WordNet’s top level.

6 COROLLARY: DIAGNOSING EXPLOITABILITY

Is there non-Euclidean inter-class structure worth exploiting for a frozen backbone, and which zero-cost metric collects it? The descriptors diagnose the first; label flatness and the training objective decide the second (full tables in Appendix A).

δ_{norm} predicts the gain within datasets. Within each dataset the Poincaré advantage tracks δ_{norm} (NC $r = -0.83$ on ImageNet), and the correlation survives partialing out dimension, baseline accuracy, family membership and spectral covariates (Appendix A.11). Without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTT (Appendix A.11). Pooled cross-dataset correlations are family-driven and are not used as evidence; on flat-label datasets no metric helps. Headline gains carry paired per-episode confidence intervals and a McNemar test (Appendix A.11).

The training objective picks the metric. The identical radial map with Euclidean distances changes nothing, so the gain is the metric itself. The split follows the objective: cosine suffices for DINOv2, Poincaré $\approx$ cosine for supervised ViTs, and Poincaré beats cosine for contrastive CLIP by up to +1.3pp even after L2 normalization (Figure 8 in Appendix A.11; per-cell values, the held-out selection and the policy comparison in Appendix A). The rule is read off the same three families it is scored on, so it describes these families rather than validating a policy.

A diagnostic, not a policy. Cosine is a safe default, and gating on δ_{norm} is the weakest policy tested (Appendix A). Gains are modest and limited to prototype-based tasks; sample-level tasks never benefit, consistent with the weak sample-level excesses of Appendix A.16. On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

7 DISCUSSION AND LIMITATIONS

What the census establishes. Foundation models organize classes into genuine, clustered structure and partially recover the human taxonomy, more so with supervision; they do not converge to one common tree. Form beyond matched nulls is widespread but not universal: near-universal in the vision backbones tested (ViTs and two ConvNets), recipe- and scale-dependent in text, always relative to a concept set. Its content is moderately shared and configuration-dependent: all recipes recover the human taxonomy partially under every non-degenerate configuration, the residual gap on ImageNet ranges from 0.13 to 0.38 with the comparison threshold, and the naive islands are a measurement artifact. Raw readings of δ , of ξ , or of tree similarity are not interpretable without the instrument; excess measures existence while raw δ_{norm} predicts exploitability, and the two dissociate in both directions (DINOv2 on ImageNet; the DBpedia embedders).

Limitations. (i) The spectrum excess certifies structure beyond second moments, star-like clustering included; against matched stars the residual depth is marginal (Appendix A.5), so the hierarchy’s reach rests on the class-subsampling control and its content on §5. (ii) The census is class-centroid geometry on six datasets; other concept sets may expose different structure. (iii) Causal-LM values (not the scale ordering) are template-conditioned. (iv) The objective–geometry link is correlational (triplet-bounded, not closed); the within-dataset δ –gain prediction rests chiefly on DINOv2’s scale range (Appendix A.11). (v) Downstream scope is prototype-based metric tasks with modest gains. (vi) The parallelogram descriptor ξ tracks norm dispersion rather than curvature (Appendix A.19) and is reported only as such. (vii) A causal LM’s class representation depends on the extraction protocol as much as on the model: left-padded batching moves $\hat{\delta}$ by up to 0.018, of the order of the L/XL excess.

ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it; all tables are generated by script from those files, and the full pipeline (feature extraction, diagnostics, nulls, calibration, tasks, statistics, figures) with fixed seeds will be released. Estimator settings are stated in §3. Computing δ_{norm} on 1000 centroids takes seconds on CPU; the full task grid runs in under 30 minutes on one consumer GPU.

THE USE OF LARGE LANGUAGE MODELS

LLM-based assistants were used, under continuous author direction and review, to draft and edit text, to write analysis scripts, and to run the experiment pipeline on the authors’ hardware. All experimental designs, results, verifications, and final claims were checked and approved by the authors, who take full responsibility for the content of this paper.

REFERENCES

- Morris Alper and Hadar Averbuch-Elor. Emergent visual-semantic hierarchies in image-text representations. In *European Conference on Computer Vision (ECCV)*, 2024.
- Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic dimension of data representations in deep neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 6109–6119, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/cfcce0621b49c983991lead4c3d4d3b6b-Abstract.html>.
- Mina Ghadimi Atigh, Julian Schoep, Erman Acar, Nanne van Noord, and Pascal Mettes. Hyperbolic image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4453–4462, June 2022.
- Ahmad Bdeir, Kristian Schwethelm, and Niels Landwehr. Fully hyperbolic convolutional neural networks for computer vision. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=ekzlhN5QNh>.
- Ines Chami, Zhitao Ying, Christopher Ré, and Jure Leskovec. Hyperbolic graph convolutional neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 4869–4880, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/0415740eaa4d9decbc8da001d3fd805f-Abstract.html>.
- Weize Chen, Xu Han, Yankai Lin, Hexu Zhao, Zhiyuan Liu, Peng Li, Maosong Sun, and Jie Zhou. Fully hyperbolic neural networks. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5672–5686, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.389. URL <https://aclanthology.org/2022.acl-long.389/>.
- Karan Desai, Maximilian Nickel, Tanmay Rajpurohit, Justin Johnson, and Ramakrishna Vedantam. Hyperbolic image-text representations. In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org, 2023.
- Aleksandr Ermolov, Leyla Mirvakhabova, Valentin Khrulkov, Nicu Sebe, and Ivan V. Oseledets. Hyperbolic vision transformers: Combining improvements in metric learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pp. 7399–7409. IEEE, 2022. doi: 10.1109/CVPR52688.2022.00726. URL <https://doi.org/10.1109/CVPR52688.2022.00726>.

- Hervé Fournier, Anas Ismail, and Antoine Vigneron. Computing the gromov hyperbolicity of a discrete metric space. *Information Processing Letters*, 115(6-8), 2015.
- Octavian Ganea, Gary Becigneul, and Thomas Hofmann. Hyperbolic neural networks. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/dbab2adc8f9d078009ee3fa810bea142-Paper.pdf.
- Fabian Gröger, Song Wen, and Maria Brbić. Revisiting the platonic representation hypothesis: An aristotelian view. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.14486.
- M. Gromov. *Hyperbolic Groups*, pp. 75–263. Springer New York, New York, NY, 1987. ISBN 978-1-4613-9586-7. doi: 10.1007/978-1-4613-9586-7_3. URL https://doi.org/10.1007/978-1-4613-9586-7_3.
- Neil He, Rishabh Anand, Hiren Madhu, Ali Maatouk, Smita Krishnaswamy, Leandros Tassiulas, Menglin Yang, and Rex Ying. HELM: Hyperbolic large language models via mixture-of-curvature experts. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL <https://openreview.net/forum?id=RnbJPkakkm>.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine A. Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (eds.), *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, Proceedings of Machine Learning Research, pp. 20617–20642. PMLR / OpenReview.net, 2024. URL <https://proceedings.mlr.press/v235/huh24a.html>.
- Valentin Khrulkov, Leyla Mirvakhabova, Evgeniya Ustinova, Ivan Oseledets, and Victor Lempitsky. Hyperbolic image embeddings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020.
- A. Sophia Koepke, Daniil Zverev, Shiry Ginosar, and Alexei A. Efros. Back into plato’s cave: Examining cross-modal representational convergence at scale, 2026. arXiv preprint.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey E. Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov (eds.), *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, Proceedings of Machine Learning Research, pp. 3519–3529. PMLR, 2019. URL <http://proceedings.mlr.press/v97/kornblith19a.html>.
- Bogdan Nica and Ján Špakula. Strong hyperbolicity. *Groups, Geometry, and Dynamics*, 10(3), 2016. arXiv:1408.0250.
- Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. In Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pp. 6338–6347, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/59dfa2df42d9e3d41f5b02bfc32229dd-Abstract.html>.
- Yann Ollivier. Ricci curvature of Markov chains on metric spaces. *Journal of Functional Analysis*, 256(3):810–864, 2009.
- Avik Pal, Max van Spengler, Guido Maria D’Amely di Melendugno, Alessandro Flaborea, Fabio Galasso, and Pascal Mettes. Compositional entailment learning for hyperbolic vision-language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=3i13Gev2hV>.
- Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020. doi: 10.1073/pnas.2015509117. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2015509117>.

- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. *arXiv preprint arXiv:2406.01506*, 2024.
- Phillip Pope, Chen Zhu, Ahmed Abdelkader, Micah Goldblum, and Tom Goldstein. The intrinsic dimension of images and its impact on learning. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL <https://openreview.net/forum?id=XJk19XzGq2J>.
- Rik Sarkar. Low distortion delaunay embedding of trees in hyperbolic plane. In Marc J. van Kreveld and Bettina Speckmann (eds.), *Graph Drawing - 19th International Symposium, GD 2011, Eindhoven, The Netherlands, September 21-23, 2011, Revised Selected Papers*, Lecture Notes in Computer Science, pp. 355–366. Springer, 2011. doi: 10.1007/978-3-642-25878-7_34. URL https://doi.org/10.1007/978-3-642-25878-7_34.
- Aditya Sinha, Siqi Zeng, Makoto Yamada, and Han Zhao. Learning structured representations with hyperbolic embeddings. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 91220–91259. Curran Associates, Inc., 2024. doi: 10.52202/079017-2895. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/a5d2da376bab7624b3caeb9f78fcaa2f-Paper-Conference.pdf.
- Alexandru Tifrea, Gary Bécigneul, and Octavian-Eugen Ganea. Poincaré glove: Hyperbolic word embeddings. In *International Conference on Learning Representations (ICLR)*, 2019.
- Menglin Yang, Ram Samarth B B, Aosong Feng, Bo Xiong, Jiahong Liu, Irwin King, and Rex Ying. Hyperbolic fine-tuning for large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL <https://openreview.net/forum?id=TkEdQv0bXB>.

A FULL NUMERICAL TABLES

Every table is generated by script directly from the released per-cell result files; no value is hand-transcribed.

A.1 WHAT EACH NULL PRESERVES

null	preserves	destroys
Gaussian, spectrum-matched	n, d , mean, covariance spectrum	all higher-order structure
PC-permutation	each principal component’s marginal	dependence across components
Cluster-preserving (tested, unused)	each cluster’s geometry, hub assignment	hub arrangement (invalid depth test: App. A.5)
Label shuffle (ablation)	the point cloud itself	embedding–label correspondence

Table 2: The nulls of the instrument and what each one preserves and destroys; “genuine” in the text always means beyond the first of these.

A.2 ESTIMATOR CALIBRATION

space	absolute δ	δ_{norm}
balanced binary tree (depth 10)	0.000	0.000
$\mathbb{H}^2$ ($K = -1$), region radius $R=2/4/8/16$	0.65/0.69/0.69/0.69	0.162/0.087/0.043/0.022
uniform S^{99} (chord / geodesic)	0.25/0.37	0.143/0.179
iid Gaussian ($n=1000$), $d=192, \dots, 1536$	—	0.104/0.081/0.074/0.061/0.055/0.046

Table 3: **A low raw δ is not evidence of hierarchy.** Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of $\mathbb{H}^2$, spheres stay high, and iid Gaussians drift low as d grows, the dimension confound. Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze.

A.3 MODEL PANEL

Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num_classes=0, i.e. CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 13); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

type	model	params
Supervised ViT (vision)	ViT-T (i21k)	5M
Supervised ViT (vision)	ViT-S (i21k)	22M
Supervised ViT (vision)	ViT-B (i21k)	86M
Supervised ViT (vision)	ViT-L (i21k)	307M
SSL (vision)	DINO-B	86M
SSL (vision)	DINOv2-S	22M
SSL (vision)	DINOv2-B	86M
SSL (vision)	DINOv2-L	307M
SSL (vision)	DINOv2-G	1.1B
Contrastive (vision)	CLIP-B	86M
Contrastive (vision)	CLIP-L	307M
Contrastive (vision)	SigLIP-B	86M
Causal LM	GPT-2 S	117M
Causal LM	GPT-2 M	345M
Causal LM	GPT-2 L	774M
Causal LM	GPT-2 XL	1.5B
Causal LM	Pythia-410M	410M
Causal LM	Pythia-1B	1.0B
Causal LM	Pythia-2.8B	2.8B
Causal LM	OLMo-1B	1.0B
Causal LM	OLMo-7B	7.0B
Text embedder	BGE-base	110M
Text embedder	BGE-large	335M
Text embedder	GTE-base	110M
Text embedder	GTE-large	335M
Text embedder	GTE-Qwen2-1.5B	1.5B
Text embedder	E5-base	110M
Text embedder	E5-large	335M

Table 4: The full model panel: 12 vision backbones (all in the census; the 10 with cached per-dataset features, i.e. all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, and 7 text embedders.

A.4 VISION CENSUS: EXCESS AND SIGNIFICANCE PER CELL

model	p99.9 IN	exc. DTD	exc. FMNIST	exc. MNIST	best-R (pp)	metric
ViT-T	+0.007	−0.046	+0.001	−0.033	+0.58	either
ViT-S	−0.005	−0.058	−0.050	−0.044	+0.49	either
ViT-B	−0.006	−0.063	−0.060	−0.030	+0.51	either
ViT-L	−0.014	−0.085	−0.070	−0.052	+0.87	either
DINO-B	−0.025	−0.060	−0.022	−0.045	—	—
DINOv2-S	−0.022	−0.049	−0.049	−0.020	+0.68	H
DINOv2-B	−0.026	−0.073	−0.078	−0.028	+1.17	cos
DINOv2-L	−0.023	−0.083	−0.058	−0.024	+1.84	cos
DINOv2-G	−0.020	−0.076	−0.066	−0.028	+2.54	cos
CLIP-B	−0.006	−0.065	−0.037	−0.015	+0.67	H
CLIP-L	−0.011	−0.059	−0.047	−0.017	+0.65	H
SigLIP-B	−0.009	−0.066	−0.058	−0.064	—	—

Table 5: Census columns moved out of Table 1. p99.9 IN: ImageNet excess under the supremum-robust statistic (20 replicates; Table 29); exc. DTD/FMNIST/MNIST: per-dataset excess on the remaining sets; best-R: few-shot advantage of the best zero-cost metric over Euclidean, mean over the four hierarchical datasets; metric: which of Poincaré (H) or cosine collects it (either: within 0.15pp; —: outside the 10-model task grid).

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	-0.021 (-2.7)	-0.043 (-3.7)	-0.059 (-2.0)	-0.037 (-3.9)	-0.022 (-0.7)	-0.049 (-1.4)
ViT-S	-0.020 (-2.3)	-0.033 (-4.0)	-0.096 (-3.9)	-0.052 (-4.1)	-0.069 (-2.3)	-0.058 (-1.9)
ViT-B	-0.028 (-3.0)	-0.017 (-1.9)	-0.090 (-3.1)	-0.058 (-4.8)	-0.076 (-2.6)	-0.043 (-1.4)
ViT-L	-0.030 (-4.4)	-0.048 (-6.0)	-0.098 (-3.7)	-0.077 (-7.5)	-0.085 (-2.9)	-0.064 (-1.9)
DINO-B	-0.005 (-0.6)	-0.068 (-10.2)	-0.097 (-3.2)	-0.051 (-4.9)	-0.041 (-1.3)	-0.070 (-2.5)
DINOv2-S	+0.012 (+1.1)	-0.053 (-8.9)	-0.095 (-2.2)	-0.040 (-4.2)	-0.068 (-2.3)	-0.036 (-1.2)
DINOv2-B	+0.006 (+0.6)	-0.043 (-11.3)	-0.135 (-2.7)	-0.068 (-6.6)	-0.095 (-3.3)	-0.045 (-1.4)
DINOv2-L	-0.008 (-1.1)	-0.051 (-17.0)	-0.155 (-3.4)	-0.078 (-7.0)	-0.074 (-2.6)	-0.042 (-1.2)
DINOv2-G	+0.004 (+0.5)	-0.079 (-24.2)	-0.157 (-3.4)	-0.071 (-6.4)	-0.081 (-3.0)	-0.042 (-1.3)
CLIP-B	-0.025 (-3.1)	-0.047 (-5.8)	-0.077 (-3.0)	-0.059 (-4.9)	-0.047 (-1.7)	-0.038 (-1.6)
CLIP-L	-0.021 (-2.3)	-0.043 (-5.8)	-0.083 (-3.3)	-0.049 (-3.9)	-0.066 (-2.0)	-0.024 (-0.6)
SigLIP-B	-0.019 (-1.7)	-0.042 (-6.6)	-0.091 (-4.0)	-0.057 (-4.8)	-0.074 (-2.6)	-0.070 (-1.9)

Table 6: Per-cell excess and z -score (in parentheses; $z = (\delta_{\text{real}} - \bar{\delta}_{\text{null}})/\sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{est}}^2}$, 5 null replicates). Bold: the three sign-positive cells.

model	d	$\hat{\delta}$	excess (z)	reps above
Barlow Twins (ResNet-50)	2048	0.075	-0.039 (-4.8)	20/20
BYOL (ResNet-50)	2048	0.073	-0.020 (-2.4)	20/20

Table 7: Backbones outside the ViT census, ImageNet centroid store: two ResNet-50 self-supervised models, under the spectrum null (20 replicates, percentile rank).

A.5 WHAT THE EXCESS CERTIFIES: SYNTHETIC CALIBRATION

synthetic cloud ($n=1000$, $d=768$)	$\hat{\delta}$	spectrum null A		hub-randomizing null B	
		Gaussian	Haar	Gaussian	Haar
30-cluster star, tight (0.1)	0.055	-0.115	-0.115	-0.126	-0.058
30-cluster star, mid (0.3)	0.059	-0.104	-0.103	-0.114	-0.049
30-cluster star, loose (0.6)	0.061	-0.084	-0.082	-0.090	-0.033
2-level hierarchy 6×5 (0.3)	0.044	-0.143	-0.143	-0.113	-0.112

Table 8: Calibration of the nulls on synthetic clouds (excess, means over seeds; within/between-cluster noise ratio in parentheses). A pure star (Gaussian clusters around Gaussian centers, no hierarchy) sits far below the spectrum null A under either construction (Gaussian coefficients, the paper’s; Haar-rotated coefficients, exact sample spectrum): null A certifies clustering, not depth. The hub-randomizing null B, which keeps every cluster intact and resamples only the hub configuration, also reports the star as “hierarchical”: a near-regular simplex of hubs already minimizes $\hat{\delta}$, so any hub randomization raises it. Null B is therefore not a valid depth test as constructed and is not used in the paper; a star-calibrated version (excess B relative to a matched star with the same cluster count and tightness) may still serve as a depth test, since under the Haar construction the two-level hierarchy sits $2\text{--}3\times$ below the star. That version is run on the real backbones in Table 9. Under the plain Gaussian null the mid-radius star scores -0.104 and the 6×5 hierarchy -0.14 : a star and a two-level tree differ by a fraction of either’s excess, which is why depth cannot be read from the spectrum excess alone.

model	CIFAR-100 ($K=20$ coarse)			ImageNet ($K=30$ WordNet)		
	real	star	depth (z)	real	star	depth (z)
ViT-T	-0.024	-0.030	+0.006 (+0.4)	-0.007	-0.006	-0.001 (-0.1)
ViT-S	-0.007	-0.044	+0.036 (+2.4)	-0.017	-0.008	-0.009 (-1.0)
ViT-B	+0.004	-0.056	+0.060 (+4.5)	-0.024	-0.013	-0.011 (-0.9)
ViT-L	-0.016	-0.056	+0.040 (+4.1)	-0.025	-0.013	-0.012 (-1.3)
DINO-B	-0.036	-0.043	+0.007 (+0.9)	-0.017	-0.006	-0.010 (-1.6)
DINOv2-S	-0.020	-0.027	+0.008 (+0.7)	-0.004	-0.001	-0.003 (-0.2)
DINOv2-B	+0.012	-0.027	+0.039 (+5.2)	-0.010	-0.002	-0.008 (-0.8)
DINOv2-L	+0.015	-0.027	+0.042 (+7.0)	-0.033	-0.004	-0.029 (-2.5)
DINOv2-G	-0.003	-0.030	+0.027 (+3.9)	-0.026	-0.005	-0.020 (-1.7)
CLIP-B	-0.025	-0.044	+0.019 (+1.7)	-0.014	-0.010	-0.004 (-0.4)
CLIP-L	-0.013	-0.049	+0.036 (+7.2)	-0.012	-0.012	-0.001 (-0.1)
SigLIP-B	-0.023	-0.050	+0.027 (+2.9)	-0.009	-0.010	+0.001 (+0.1)

Table 9: Star-calibrated depth test (excess B, Haar construction). For each backbone the class centroids are compared with a *matched star*: K Gaussian hubs with the real hubs’ RMS radius and, around each hub, an isotropic Gaussian cloud with that superclass’s real within-cluster RMS spread (same n , d , K and cluster sizes; 3 star seeds). “real” and “star” are excesses of $\hat{\delta}$ over the hub-randomizing null (each cluster kept intact, hubs replaced by a Haar-rotated sample with the hubs’ exact spectrum, 10 replicates), which has a small star bias of its own; “depth” is their difference (real minus star; negative = more tree-like than a matched star), with z against the combined star, null and estimator spread. On ImageNet only DINOv2-L exceeds its matched star beyond two spreads ($z = -2.5$; DINOv2-G -1.7 , all others $|z| \leq 1.6$); on CIFAR-100 every backbone is *less* tree-like than its matched star. The excess of §4 is therefore accounted for by clustered, star-like organization; residual depth is at most marginal.

model	CIFAR-100 images ($K=20$)			ImageNet images ($K=30$)		
	exc. (z)	depth (z)	$\hat{\delta}$ nat./Eucl.	exc. (z)	depth (z)	$\hat{\delta}$ nat./Eucl.
MERU ViT-S	-0.040 (-3.0)	+0.023 (+1.7)	0.163/0.163	-0.024 (-2.5)	-0.004 (-0.4)	0.128/0.128
CLIP ViT-S	-0.059 (-4.9)	-0.001 (-0.1)	0.140/0.141	-0.025 (-2.7)	-0.005 (-0.5)	0.151/0.126
MERU ViT-B	-0.042 (-3.4)	+0.023 (+2.5)	0.156/0.156	-0.022 (-2.4)	+0.000 (+0.0)	0.129/0.129
CLIP ViT-B	-0.056 (-4.4)	+0.010 (+0.9)	0.146/0.141	-0.025 (-2.3)	-0.001 (-0.1)	0.151/0.124
MERU ViT-L	-0.023 (-1.5)	+0.043 (+3.1)	0.170/0.171	-0.021 (-2.2)	+0.004 (+0.4)	0.128/0.128
CLIP ViT-L	-0.037 (-2.9)	+0.025 (+2.1)	0.170/0.155	-0.025 (-2.5)	-0.003 (-0.3)	0.150/0.123

Table 10: Hyperbolic-backbone control. MERU (Desai et al., 2023) embeds images on the Lorentz hyperboloid with an entailment objective; its released Euclidean twin (a CLIP baseline: same ViT, same RedCaps data and recipe, minus the hyperbolic lift and entailment loss) differs only in geometry. Both pass through the paper’s instrument unchanged (class centroids of the projected embeddings; “exc.” is the excess over the spectrum-matched null, “depth” the star-calibrated test of Table 9, “nat.” the Gromov $\hat{\delta}$ computed with the model’s own metric: Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP). Training in hyperbolic space changes nothing at the class level: MERU shows the same clustering excess as its twin, no residual depth beyond a matched star on either dataset, and its Lorentz-metric $\hat{\delta}$ is indistinguishable from the Euclidean one. MERU’s hierarchy is a generic $\rightarrow$ specific (text $\supset$ image) partial order, not a class taxonomy; the class-tree depth that no standard backbone shows is not present in a hyperbolic one either. The synthetic two-level tree of Table 8 remains the existence proof that the depth test fires when depth is present.

A.6 TEXT CENSUS

model	type	d	δ_{norm}	null	excess	z
GPT-2 S	causal LM	768	0.131±0.007	0.083±0.002	+0.048	+6.5
GPT-2 M	causal LM	1024	0.091±0.004	0.103±0.002	-0.013	-2.7
GPT-2 L	causal LM	1280	0.095±0.004	0.129±0.006	-0.034	-5.0
GPT-2 XL	causal LM	1600	0.083±0.005	0.131±0.007	-0.048	-5.4
Pythia-410M	causal LM	1024	0.098±0.005	0.136±0.005	-0.037	-5.4
Pythia-1B	causal LM	2048	0.096±0.004	0.131±0.004	-0.035	-6.2
Pythia-2.8B	causal LM	2560	0.098±0.004	0.139±0.008	-0.041	-4.6
OLMo-1B	causal LM	2048	0.100±0.005	0.101±0.002	-0.001	-0.2
OLMo-7B	causal LM	4096	0.093±0.005	0.108±0.002	-0.015	-2.8
BGE-base	embedder	768	0.121±0.006	0.117±0.005	+0.005	+0.6
BGE-large	embedder	1024	0.121±0.008	0.120±0.003	+0.001	+0.1
GTE-base	embedder	768	0.124±0.011	0.113±0.003	+0.011	+0.9
GTE-large	embedder	1024	0.120±0.012	0.116±0.003	+0.004	+0.4
E5-base	embedder	768	0.115±0.007	0.112±0.005	+0.003	+0.3
E5-large	embedder	1024	0.114±0.010	0.113±0.003	+0.001	+0.1
GTE-Qwen2-1.5B	embedder	1536	0.107±0.004	0.126±0.004	-0.019	-3.1

Table 11: Text models under the calibrated protocol (1000 ImageNet class prompts): δ_{norm} , spectrum-matched null (3 reps) and excess.

model	padding-free re-extraction, 200 replicates				original extraction, 3 replicates (Table 11)		
	$\hat{\delta}$	excess	$r/200$	p	$\hat{\delta}$	excess	$\Delta\hat{\delta}$
GPT-2 S	0.128	-0.013	183	.090	0.131	+0.048	-0.003
GPT-2 M	0.082	-0.009	173	.139	0.091	-0.013	-0.009
GPT-2 L	0.082	-0.039	200	.005	0.095	-0.034	-0.013
GPT-2 XL	0.072	-0.047	200	.005	0.083	-0.048	-0.011
Pythia-410M	0.098	-0.035	200	.005	0.098	-0.037	+0.000
Pythia-1B	0.096	-0.031	200	.005	0.096	-0.035	+0.000
Pythia-2.8B	0.098	-0.035	200	.005	0.098	-0.041	-0.000
OLMo-1B	0.082	-0.038	200	.005	0.100	-0.001	-0.018
OLMo-7B	—	—	—	—	0.093	-0.015	—
BGE-base	0.121	+0.005	23	.886	0.121	+0.005	+0.000
BGE-large	0.121	+0.003	53	.736	0.121	+0.001	-0.000
GTE-base	0.124	+0.012	2	.990	0.124	+0.011	+0.000
GTE-large	0.120	+0.007	11	.945	0.120	+0.004	-0.000
GTE-Qwen2-1.5B	0.122	+0.003	63	.687	0.107	-0.019	+0.015
E5-base	0.115	+0.004	39	.806	0.115	+0.003	-0.000
E5-large	0.114	+0.005	28	.861	0.114	+0.001	+0.000

Table 12: **The text census of record: padding-free extraction scored against 200 spectrum-null replicates.** r = replicates above the real value, p the left-tail add-one p -value (genuine = $p \leq 0.05$, $r \geq 191$); OLMo-7B not re-run (exceeds the local GPU). Raw $\hat{\delta}$ differs from the original extraction because the embeddings were re-extracted, not because of the replicate count; $\Delta\hat{\delta}$ is that extraction-to-extraction difference. Sign agreement with the 3-replicate original: 13/15. Three verdicts change between extractions, each with a known cause: GPT-2 S (above null in the left-padded original; Table 13), OLMo-1B (the original used a different checkpoint revision), and GTE-Qwen2 (genuine in the original with 3 replicates, -0.019; here excess +0.003, $r = 63$, $p = .687$; not used as evidence).

batch size	GPT-2 M		OLMo-1B	
	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$
1 (no padding)	0.0816	0.0817	0.0817	0.0817
16	0.0996	0.0997	0.0817	0.0816
32	0.0905	0.0905	0.0817	0.0817

Table 13: Where extraction-to-extraction variability comes from. Same 1000 prompts, same model; only batch size and precision vary. Precision is irrelevant (fp16 equals fp32 to four decimals). For GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves $\hat{\delta}$ by up to 0.018; OLMo handles left padding correctly and is invariant. Padding-free (batch-size-1) extraction is therefore the protocol of Table 12.

A.7 CLASS-COUNT VS HIERARCHY DEPTH

model	C	random subsets (span the hierarchy)			WordNet-coherent (siblings)		
		δ_{norm}	excess	NC adv	δ_{norm}	excess	NC adv
DINOv2-L	10	0.030	-0.121	+1.04	0.068	-0.096	+0.08
DINOv2-L	20	0.040	-0.157	+1.32	0.080	-0.118	+0.12
DINOv2-L	50	0.049	-0.124	+1.58	0.087	-0.093	+0.53
DINOv2-L	100	0.060	-0.081	+1.46	0.085	-0.061	+0.77
DINOv2-L	200	0.062	-0.054	+1.45	0.087	-0.034	+0.59
DINOv2-L	500	0.061	-0.026	+1.01	0.070	-0.021	+0.82
DINOv2-L	1000	0.059	-0.016	+0.57	—	—	—
DINOv2-G	10	0.029	-0.123	+0.80	0.072	-0.094	-0.56
DINOv2-G	20	0.042	-0.155	+1.30	0.077	-0.120	-0.18
DINOv2-G	50	0.060	-0.111	+1.51	0.090	-0.092	+0.26
DINOv2-G	100	0.068	-0.073	+1.30	0.090	-0.056	+0.36
DINOv2-G	200	0.076	-0.039	+1.28	0.090	-0.033	+0.27
DINOv2-G	500	0.082	-0.003	+0.49	0.077	-0.013	+0.22
DINOv2-G	1000	0.076	+0.003	-0.13	—	—	—
CLIP-L	10	0.071	-0.093	-0.08	0.099	-0.097	-0.36
CLIP-L	20	0.090	-0.099	+0.20	0.129	-0.067	-0.24
CLIP-L	50	0.110	-0.076	+0.34	0.147	-0.051	-0.46
CLIP-L	100	0.115	-0.051	+0.21	0.130	-0.040	-0.31
CLIP-L	200	0.113	-0.038	+0.00	0.124	-0.025	-0.35
CLIP-L	500	0.113	-0.031	-0.27	0.114	-0.027	-0.42
CLIP-L	1000	0.112	-0.023	-0.53	—	—	—

Table 14: Class-count vs hierarchy-depth deconfound on ImageNet (advantages in pp; means over subset seeds; $C \leq 30$ uses exact quadruple enumeration).

A.8 TREE-MAP METRIC AND LINKAGE CONTROLS

Threshold sensitivity of the selection criterion of §5: the selected configuration is stable for degeneracy bounds in $[0.45, 0.50]$ on ImageNet and $[0.40, 0.50]$ on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

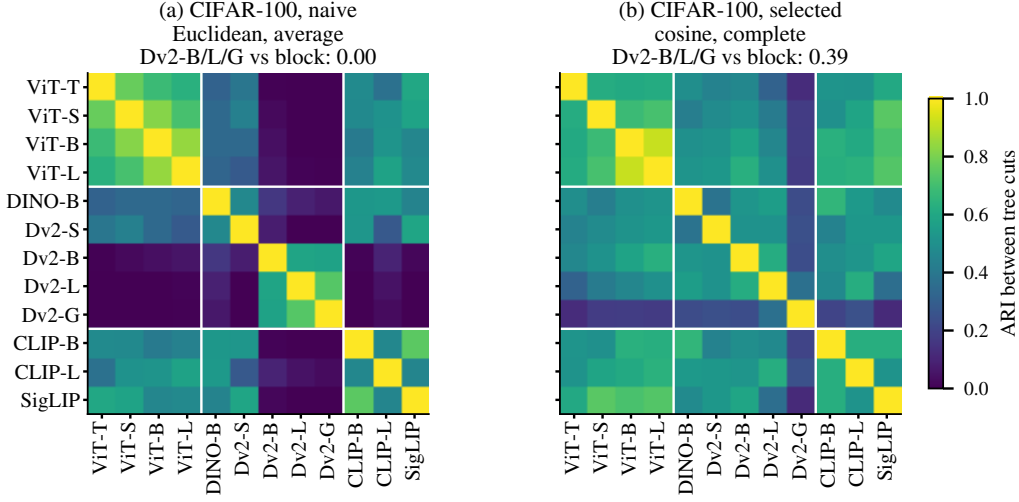


Figure 7: **The island is an artifact on CIFAR-100 too.** Same reading as Figure 6 on the 20 CIFAR-100 superclass cuts: naive (a) vs criterion-selected (b).

dataset	configuration	max frac.	CPCC	Dv2-B/L/G vs block	Dv2-S+DINO-B vs block	within block
imagenet	euclid-average (deg.)	0.97	0.761	0.03	0.08	0.64
imagenet	euclid-complete (deg.)	0.83	0.606	0.03	0.17	0.44
imagenet	euclid-ward	0.48	0.464	0.15	0.38	0.53
imagenet	cosine-average (selected)	0.40	0.691	0.38	0.35	0.48
imagenet	cosine-complete	0.32	0.539	0.13	0.17	0.50
imagenet	cosine-ward	0.37	0.464	0.17	0.39	0.55
cifar100	euclid-average (deg.)	0.80	0.798	0.00	0.41	0.57
cifar100	euclid-complete (deg.)	0.66	0.722	0.17	0.40	0.54
cifar100	euclid-ward	0.33	0.534	0.37	0.65	0.63
cifar100	cosine-average (deg.)	0.53	0.777	0.30	0.54	0.58
cifar100	cosine-complete (selected)	0.39	0.698	0.39	0.51	0.63
cifar100	cosine-ward	0.14	0.584	0.56	0.62	0.61

Table 15: Tree-map configurations with the two quantities that drive the selection criterion of §5: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix. Configurations with max-cluster fraction > 0.5 are excluded; among the rest, highest CPCC selects the marked row. Mean pairwise ARIs at the reference cut shown for all configurations.

model	euclid-avg (naive, degen.)	euclid-ward	cosine-avg (degen. C100)	cosine-complete ← sel.	cosine-ward
ViT-T	0.31	0.38	0.29	0.33	0.33
ViT-S	0.35	0.56	0.37	0.47	0.56
ViT-B	0.39	0.61	0.43	0.45	0.57
ViT-L	0.38	0.55	0.49	0.48	0.60
DINO-B	0.22	0.48	0.28	0.38	0.54
DINOv2-S	0.12	0.51	0.32	0.33	0.49
DINOv2-B	0.02	0.47	0.35	0.42	0.51
DINOv2-L	0.01	0.32	0.25	0.50	0.48
DINOv2-G	0.00	0.19	0.08	0.20	0.52
CLIP-B	0.18	0.48	0.18	0.37	0.48
CLIP-L	0.35	0.52	0.32	0.41	0.54
SigLIP-B	0.22	0.49	0.22	0.37	0.52

Table 16: Recovery of the true CIFAR-100 superclasses (ARI of the 20-cluster cut) per model and configuration, the quantity behind the corrected picture of §5. Degenerate configurations marked; the criterion-selected configuration is cosine-complete.

A.9 TASK ACCURACIES (REGENERATED)

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	68.1/67.2	29.3/28.2	87.5/87.4	63.7/62.5	78.5/78.4	75.9/74.0
ViT-S	77.7/77.5	74.9/74.6	93.4/93.2	69.3/68.5	80.7/80.4	84.8/84.1
ViT-B	81.6/81.7	77.3/77.0	94.5/94.0	70.8/70.8	81.3/81.6	81.6/80.5
ViT-L	86.1/86.2	85.3/85.2	97.5/97.5	71.8/71.5	83.7/83.5	84.2/83.6
DINOv2-S	74.7/74.4	73.4/74.4	95.0/95.2	73.4/73.4	82.2/82.2	83.4/81.9
DINOv2-B	78.9/78.9	87.4/87.9	97.1/97.4	74.2/74.7	83.2/82.8	80.6/78.8
DINOv2-L	79.8/80.4	85.6/87.7	97.9/98.3	74.4/75.2	84.4/84.5	74.1/72.7
DINOv2-G	78.4/78.3	89.8/91.0	97.8/98.2	72.9/73.1	85.9/85.8	79.3/78.6
CLIP-B	61.4/59.9	65.6/65.1	87.9/88.3	63.7/63.2	75.9/75.1	70.1/66.6
CLIP-L	75.0/74.5	74.6/74.5	95.9/95.9	68.6/69.0	76.2/75.9	85.2/86.1

Table 17: Nearest-centroid accuracy (%), Euclidean/Poincaré, regenerated from the audited reruns.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	97.5/97.6	62.4/63.1	87.6/88.1	83.3/83.9	81.6/81.9	74.3/74.6
ViT-S	98.6/98.7	92.2/92.4	92.0/92.5	85.4/86.0	83.7/83.5	80.5/81.4
ViT-B	98.9/99.0	92.5/92.7	92.7/93.0	85.4/86.2	83.4/83.3	76.8/77.6
ViT-L	99.4/99.4	94.8/95.5	95.0/95.8	85.6/86.5	84.4/84.7	80.8/81.6
DINOv2-S	97.4/97.9	94.3/95.1	92.9/93.5	89.8/90.5	85.4/85.5	80.2/80.8
DINOv2-B	97.0/97.8	96.5/97.2	94.7/95.4	89.2/90.3	85.9/85.9	76.6/77.1
DINOv2-L	96.2/97.1	96.1/97.1	94.1/95.1	87.1/88.4	87.8/87.9	70.5/71.1
DINOv2-G	96.2/97.1	95.9/97.2	92.3/93.8	87.2/88.5	89.0/89.2	72.9/73.8
CLIP-B	96.8/96.8	87.7/88.9	89.5/90.1	82.2/83.1	78.8/78.9	70.9/69.7
CLIP-L	98.4/98.6	90.0/90.8	92.4/93.0	85.2/86.2	76.8/77.3	83.7/84.7

Table 18: 5-way 5-shot accuracy (%), Euclidean/Poincaré, 1000 paired episodes.

A.10 METRIC CONTROLS

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	-0.01	+1.01	+0.66	-0.23	+0.21	+0.64
ViT-S	+0.17	+0.59	+0.56	-0.03	+0.28	+0.34
ViT-B	+0.20	+0.56	+0.49	+0.28	+0.03	+0.19
ViT-L	+0.11	+1.17	+1.25	+0.13	+0.00	+0.08
DINOv2-S	+0.68	+0.80	-0.36	+0.30	+0.02	-0.07
DINOv2-B	+1.40	+1.11	+0.27	+1.48	+0.21	+0.17
DINOv2-L	+2.13	+1.80	+0.92	+2.40	+0.29	+0.25
DINOv2-G	+2.57	+2.44	+2.51	+2.63	+0.38	+0.30
CLIP-B	-0.04	-0.03	-0.19	-0.23	+0.06	+0.04
CLIP-L	+0.01	+0.12	+0.16	-0.08	+0.12	+0.08

Table 19: Few-shot advantage of cosine over Euclidean (pp).

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	+0.01	+0.45	+0.19	+0.32	+0.10	+0.52
ViT-S	-0.02	+0.16	+0.27	+0.25	+0.08	+0.65
ViT-B	-0.04	+0.12	+0.18	+0.19	+0.09	+0.51
ViT-L	-0.03	+0.31	+0.35	+0.16	+0.17	+0.42
DINOv2-S	+0.03	+0.15	+0.13	+0.04	+0.07	+0.45
DINOv2-B	+0.04	+0.09	+0.03	+0.01	+0.04	+0.48
DINOv2-L	+0.02	+0.09	-0.00	+0.03	+0.03	+0.43
DINOv2-G	+0.03	+0.13	-0.02	+0.01	+0.04	+0.56
CLIP-B	-0.00	+0.44	+0.20	+0.36	+0.17	+0.23
CLIP-L	+0.00	+0.39	+0.25	+0.36	+0.33	+0.58

Table 20: Few-shot advantage of the radial-rescaling control over Euclidean (pp): the map without the metric changes nothing.

A.11 STATISTICS

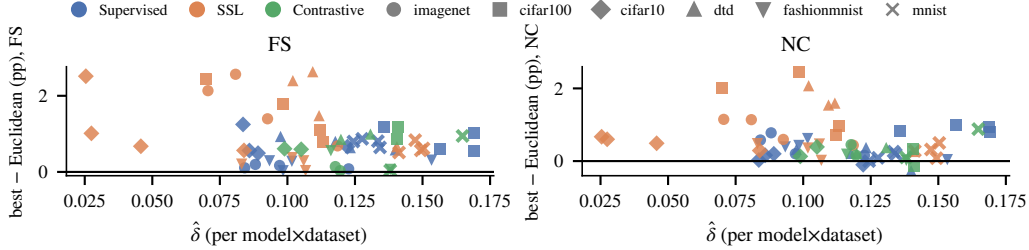


Figure 8: **More tree-like cells carry larger zero-cost metric gains.** Best zero-cost metric minus Euclidean per cell vs δ_{norm} (FS left, NC right). Within-dataset correlations -0.60 to -0.83 (FS) and -0.61 to -0.86 (NC); pooled correlations are family-driven and not used as evidence (Appendix A.11).

measure	raw	null mean	$\tau_{0.05}$	calibrated	null-centered	pairs $p < 0.05$
mutual-kNN ($k=10$), Euclidean	0.472	0.010	0.015	0.464	0.462	66/66
mutual-kNN, Poincaré	0.425	0.010	0.019	0.414	0.415	66/66
linear CKA, Euclidean	0.633	0.108	0.111	0.577	0.524	66/66
linear CKA, Poincaré	0.640	0.108	0.111	0.586	0.532	66/66

Table 21: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with $K=200$ permutations of the class correspondence and $\alpha=0.05$, scalar (no layer search): $\tau_{0.05}$ is the $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9), $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (K+1)$ (eq. 10), calibrated = $\max\{(\text{obs} - \tau_{0.05}) / (1 - \tau_{0.05}), 0\}$ (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level $k/(n-1) = 10/999$.

quantity	dataset	pooled r	demeaned r	ViT	DINOv2	CLIP
NC H-R	ImageNet	-0.83	-0.73	-0.99	-0.77	(-1)
	CIFAR-100	-0.87	-0.45	-0.68	-0.33	(-1)
	CIFAR-10	-0.63	-0.31	+0.01	-0.98	(+1)
	DTD	-0.46	-0.57	-0.78	-0.76	(+1)
FS H-R	ImageNet	-0.57	-0.73	-0.04	-0.97	(-1)
	CIFAR-100	-0.67	-0.63	-0.41	-0.96	(+1)
	CIFAR-10	-0.78	-0.64	-0.22	-0.79	(-1)
	DTD	-0.71	-0.86	-0.95	-0.98	(+1)
FS best-R	ImageNet	-0.64	-0.71	-0.57	-0.90	(-1)
	CIFAR-100	-0.83	-0.82	-0.58	-0.96	(+1)
	CIFAR-10	-0.63	-0.55	-0.26	-0.65	(-1)
	DTD	-0.60	-0.70	-0.95	-0.88	(+1)

Table 22: Family-identity control for the δ_{norm} -gain correlations (within dataset, 10 backbones = 3 families with nested scales). Demeaned: both variables centered within family before correlating; ViT/DINOv2: within-family correlations (4 models each); CLIP has only two models, so its within-family value (parenthesized sign) is not informative. Without DINOv2 (six backbones) the NC correlation holds on ImageNet and CIFAR-100 ($-0.87/-0.65$) but not on CIFAR-10/DTD ($+0.23/-0.34$). The within-dataset prediction survives the family control; pooled cross-dataset correlations of the best-metric advantage do not ($-0.31 \rightarrow -0.18$ over all 60 cells, $-0.33 \rightarrow -0.04$ on hierarchical cells) and are therefore family-driven.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	10^{-22} (-)	10^{-5} (-)	0.19 (-)	0.03 (-)	0.38 (-)	10^{-13} (-)
ViT-S	0.09 (-)	0.10 (-)	0.02 (-)	0.10 (-)	0.08 (-)	10^{-4} (-)
ViT-B	0.21 (+)	0.07 (-)	10^{-5} (-)	1.00 (-)	0.08 (+)	10^{-7} (-)
ViT-L	0.58 (+)	0.59 (-)	0.76 (+)	0.69 (-)	0.12 (-)	0.00 (-)
DINOv2-S	0.00 (-)	10^{-6} (+)	0.14 (+)	1.00 (+)	1.00 (-)	10^{-11} (-)
DINOv2-B	0.57 (+)	10^{-4} (+)	10^{-4} (+)	0.31 (+)	0.01 (-)	10^{-16} (-)
DINOv2-L	10^{-14} (+)	10^{-38} (+)	10^{-8} (+)	0.11 (+)	0.40 (+)	10^{-12} (-)
DINOv2-G	0.10 (-)	10^{-16} (+)	10^{-13} (+)	0.81 (+)	0.72 (-)	0.00 (-)
CLIP-B	10^{-40} (-)	0.07 (-)	10^{-4} (+)	0.46 (-)	10^{-4} (-)	10^{-45} (-)
CLIP-L	10^{-8} (-)	0.66 (-)	0.59 (-)	0.45 (+)	0.09 (-)	10^{-7} (+)

Table 23: McNemar p -values for NC test-set decisions, Poincaré vs Euclidean (order of magnitude when $p < 10^{-3}$; sign of the NC advantage in parentheses, so significance is readable with its direction).

task	dataset	r	CI95
NC	imagenet	-0.83	[-0.97, -0.65]
NC	cifar100	-0.87	[-0.97, -0.78]
NC	cifar10	-0.63	[-0.97, -0.06]
NC	dtd	-0.46	[-0.90, +0.22]
FS	imagenet	-0.57	[-0.91, +0.14]
FS	cifar100	-0.67	[-0.95, -0.15]
FS	cifar10	-0.78	[-0.93, -0.23]
FS	dtd	-0.71	[-0.94, -0.32]
NC	POOLED(cluster-bs)	-0.45	[-0.60, -0.36]
FS	POOLED(cluster-bs)	-0.18	[-0.50, +0.12]

Table 24: Bootstrap CI95 for the correlations between δ_{norm} and the projection advantage (within-dataset: 10k resamples of the 10 models; pooled: cluster bootstrap over datasets).

policy (test advantage over Euclidean, pp)	all 60 cells	hierarchical (40)
always cosine	+0.23	+0.28
objective rule (zero validation)	+0.28	+0.41
δ -gated objective rule	+0.08	+0.13
validation-picked per cell	+0.63	+0.78
test oracle (reference)	+0.64	+0.78

Table 25: Metric-selection policies evaluated on held-out episodes (500 validation / 500 test per cell, paper protocol). Validation picking retains the oracle advantage; the zero-validation objective rule modestly beats always-cosine.

NC policy (advantage over Euclidean, pp)	all 60	flat (20)	hierarchical (40)
always cosine	+0.48	+0.24	+0.59
always Poincaré	-0.24	-0.70	-0.00
gated rule (flat→R)	+0.34	+0.00	+0.51
δ -gated rule	+0.19	+0.00	+0.29

Table 26: NC metric-selection policies. The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere.

A.12 PROJECTION-TARGET SENSITIVITY

target t	0.25	0.35	0.41	0.45	0.50	0.58	0.71	0.95
FS adv. mean (pp)	+0.88	+0.87	+0.87	+0.86	+0.84	+0.82	+0.76	+0.56
NC adv. mean (pp)	-0.30	-0.25	-0.23	-0.21	-0.17	-0.12	-0.01	+0.15
% cells FS > 0	97	100	100	100	100	100	100	100

Table 27: Sensitivity to the projection target t (36 hierarchical cells). The headline $t = 1/\sqrt{2} \approx 0.71$ was fixed a priori and is conservative for FS.

A.13 ESTIMATOR ROBUSTNESS

model	data	excess vs quadruple budget ($10^4 \dots 2 \times 10^6$)						p99.9 @ 5×10^5
ViT-L	IN	-0.026	-0.024	-0.027	-0.030	-0.034	-0.035	-0.018
DINOv2-L	IN	-0.027	-0.019	-0.024	-0.011	-0.004	-0.005	-0.024
CLIP-B	IN	-0.008	-0.012	-0.023	-0.025	-0.025	-0.031	-0.007
ViT-L	C100	-0.045	-0.047	-0.047	-0.049	-0.052	-0.053	-0.033
DINOv2-L	C100	-0.071	-0.055	-0.054	-0.050	-0.052	-0.055	-0.069
CLIP-B	C100	-0.045	-0.047	-0.050	-0.049	-0.052	-0.051	-0.030
ViT-L	DTD	-0.064	-0.070	-0.076	-0.083	-0.084	-0.084	-0.054
DINOv2-L	DTD	-0.068	-0.072	-0.075	-0.083	-0.084	-0.084	-0.064
CLIP-B	DTD	-0.048	-0.055	-0.059	-0.065	-0.066	-0.066	-0.035
Centroid bootstrap (30 resamples of the per-class images): s.d. of the excess								
ViT-L	C100	excess	-0.0493	bootstrap s.d.	0.0029	(30 resamples, 500 images/class)		
DINOv2-L	C100	excess	-0.0493	bootstrap s.d.	0.0022	(30 resamples, 500 images/class)		
DINOv2-G	C100	excess	-0.0778	bootstrap s.d.	0.0020	(30 resamples, 500 images/class)		
CLIP-B	C100	excess	-0.0476	bootstrap s.d.	0.0017	(30 resamples, 500 images/class)		
ViT-L	DTD	excess	-0.0803	bootstrap s.d.	0.0049	(30 resamples, 80 images/class)		
DINOv2-L	DTD	excess	-0.0878	bootstrap s.d.	0.0048	(30 resamples, 80 images/class)		
DINOv2-G	DTD	excess	-0.0863	bootstrap s.d.	0.0051	(30 resamples, 80 images/class)		
CLIP-B	DTD	excess	-0.0581	bootstrap s.d.	0.0054	(30 resamples, 80 images/class)		

Table 28: Estimator robustness. Top: on CIFAR-100 and DTD the excess is stable once the quadruple budget reaches 10^5 (the paper uses 5×10^5); on ImageNet ($C=1000$) every sign holds at every budget but magnitudes drift by up to 0.02 (DINOv2-L toward its null, ViT-L and CLIP-B away from it), so ImageNet excess magnitudes carry a budget caveat. The 99.9th-percentile variant of the four-point statistic, robust to the supremum, is negative wherever the supremum excess is. Bottom: resampling the per-class images that form each centroid (reported on CIFAR-100 and DTD) moves the excess by an order of magnitude less than its size.

model	p99.9 $\hat{\delta}$	excess	z	null reps above real
ViT-T	0.100	+0.0069	+1.3	1/20
ViT-S	0.079	-0.0048	-1.1	17/20
ViT-B	0.081	-0.0065	-1.8	20/20
ViT-L	0.069	-0.0139	-4.3	20/20
DINO-B	0.054	-0.0247	-10.9	20/20
DINOv2-S	0.053	-0.0220	-16.1	20/20
DINOv2-B	0.035	-0.0256	-33.1	20/20
DINOv2-L	0.032	-0.0235	-42.2	20/20
DINOv2-G	0.032	-0.0203	-37.9	20/20
CLIP-B	0.087	-0.0061	-1.6	19/20
CLIP-L	0.076	-0.0114	-3.1	20/20
SigLIP-B	0.076	-0.0090	-2.7	20/20

Table 29: The ImageNet census under the supremum-robust statistic: 99.9th percentile of the four-point defect, 20 spectrum-null replicates, with the percentile rank (how many replicates exceed the real value; 20/20 = below every replicate). The ordering shifts relative to the supremum: DINO/DINOv2 carry the most budget-robust excess, while ViT-T’s supremum excess does not survive (+0.007, above 19/20 replicates), consistent with the ordering of the norm-structure descriptor ξ (Table 39). Large $|z|$ should be read as “below every replicate”, not as a Gaussian tail probability.

	IN			C100			C10			DTD			FMNIST			MNIST		
model	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p
ViT-T	-0.020	199	.010	-0.038	200	.005	-0.042	191	.050	-0.046	200	.005	+0.001	95	.527	-0.033	178	.114
ViT-S	-0.022	200	.005	-0.031	199	.010	-0.081	200	.005	-0.058	200	.005	-0.050	198	.015	-0.044	186	.075
ViT-B	-0.025	200	.005	-0.016	187	.070	-0.073	200	.005	-0.063	200	.005	-0.060	199	.010	-0.030	181	.100
ViT-L	-0.030	200	.005	-0.044	200	.005	-0.088	200	.005	-0.085	200	.005	-0.070	200	.005	-0.052	196	.025
DINO-B	-0.003	162	.194	-0.064	200	.005	-0.085	200	.005	-0.060	200	.005	-0.022	164	.184	-0.045	193	.040
DINOv2-S	+0.011	0	1	-0.052	200	.005	-0.095	200	.005	-0.049	200	.005	-0.049	196	.025	-0.020	154	.234
DINOv2-B	+0.007	0	1	-0.042	200	.005	-0.135	200	.005	-0.073	200	.005	-0.078	200	.005	-0.028	162	.194
DINOv2-L	-0.008	200	.005	-0.051	200	.005	-0.153	200	.005	-0.083	200	.005	-0.058	199	.010	-0.024	163	.189
DINOv2-G	+0.005	1	.995	-0.079	200	.005	-0.155	200	.005	-0.076	200	.005	-0.066	200	.005	-0.028	162	.194
CLIP-B	-0.025	200	.005	-0.044	200	.005	-0.062	200	.005	-0.065	200	.005	-0.037	177	.119	-0.015	137	.318
CLIP-L	-0.018	200	.005	-0.039	200	.005	-0.072	200	.005	-0.059	200	.005	-0.047	191	.050	-0.017	144	.284
SigLIP-B	-0.016	199	.010	-0.038	200	.005	-0.080	200	.005	-0.066	200	.005	-0.058	200	.005	-0.064	200	.005

Table 30: **The census of record: excess, percentile rank and left-tail p per cell.** Excess over the spectrum-matched null; r = number of the 200 null replicates above the real value; $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; genuine = $p \leq 0.05$ ($r \geq 191$). Genuine cells: 55/72 overall, 43/48 on the hierarchical-label datasets. Secondary z summary (excess over the combined null and estimator s.d.): $|z| \geq 2$ in 49/72 cells, $z \leq -3$ in 32/48 hierarchical cells. Sign agreement with the 5-replicate census (Table 6): 71/72. Every cell is computed on the census cache (one centroid source; the ImageNet cache regenerated with the original extraction pipeline).

	IN		C100		C10		DTD		FMNIST		MNIST	
model	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r
ViT-T	+0.007 [†]	1	-0.008	15	-0.051	20	-0.024	20	-0.012	11	-0.041	19
ViT-S	-0.005 [†]	18	-0.005	15	-0.090	20	-0.035	20	-0.060	20	-0.051	20
ViT-B	-0.006 [†]	20	+0.001	12	-0.083	20	-0.042	20	-0.069	20	-0.041	20
ViT-L	-0.014 [†]	20	-0.029	20	-0.096	20	-0.054	20	-0.079	20	-0.060	20
DINO-B	-0.025	20	-0.041	20	-0.094	20	-0.041	20	-0.035	18	-0.058	20
DINOv2-S	-0.022	20	-0.047	20	-0.095	20	-0.032	20	-0.061	20	-0.029	17
DINOv2-B	-0.026	20	-0.057	20	-0.134	20	-0.057	20	-0.089	20	-0.036	18
DINOv2-L	-0.023	20	-0.067	20	-0.154	20	-0.065	20	-0.068	20	-0.032	19
DINOv2-G	-0.020	20	-0.073	20	-0.155	20	-0.061	20	-0.075	20	-0.037	19
CLIP-B	-0.006	19	-0.025	20	-0.071	20	-0.035	20	-0.042	19	-0.028	16
CLIP-L	-0.011	20	-0.026	20	-0.080	20	-0.039	20	-0.058	20	-0.023	15
SigLIP-B	-0.009	20	-0.029	20	-0.088	20	-0.039	20	-0.068	20	-0.067	20

Table 31: The full vision p99.9 census with 20 spectrum-null replicates: excess and percentile rank r (number of the 20 null replicates above the real value; 20 = below every replicate). Agreement with the 5-replicate census on the 68 directly comparable cells: sign 64/68, significance at $|z| \geq 2$ 54/68; 56/72 cells lie below every replicate. [†]ImageNet rows for the supervised ViTs use the precomputed centroid store (precomputed centroid store, whose supervised-ViT centroids differ slightly from the census cache) and are not directly comparable to Table 6. The distributional statistic is stricter on supervised ViTs and stronger on DINO/DINOv2 than the supremum (Table 29).

model	IN			C100			C10		
	Gauss.	Haar	PC-perm.	Gauss.	Haar	PC-perm.	Gauss.	Haar	PC-perm.
ViT-T	-0.005	-0.014	-0.038	-0.044	-0.031	-0.055	-0.057	-0.027	-0.050
ViT-S	-0.023	-0.025	-0.052	-0.034	-0.025	-0.049	-0.098	-0.063	-0.090
ViT-B	-0.027	-0.025	-0.054	-0.018	-0.011	-0.039	-0.090	-0.054	-0.078
ViT-L	-0.028	-0.029	-0.040	-0.047	-0.034	-0.051	-0.104	-0.066	-0.096
DINO-B	-0.005	-0.002	-0.008	-0.068	-0.056	-0.068	-0.100	-0.070	-0.097
DINOv2-S	+0.010	+0.015	+0.013	-0.050	-0.029	-0.051	-0.093	-0.083	-0.098
DINOv2-B	+0.004	+0.015	+0.006	-0.042	-0.007	-0.041	-0.132	-0.120	-0.124
DINOv2-L	-0.013	+0.000	-0.011	-0.053	-0.008	-0.044	-0.153	-0.136	-0.122
DINOv2-G	-0.000	+0.016	+0.001	-0.080	-0.032	-0.074	-0.155	-0.138	-0.110
CLIP-B	-0.028	-0.028	-0.048	-0.046	-0.036	-0.053	-0.079	-0.045	-0.075
CLIP-L	-0.016	-0.017	-0.029	-0.042	-0.031	-0.052	-0.087	-0.049	-0.082
SigLIP-B	-0.019	-0.020	-0.044	-0.042	-0.032	-0.046	-0.096	-0.056	-0.081

Table 32: Excess under three constructions of the spectrum-matched null (10 replicates each): Gaussian coefficients (the paper’s), Haar-rotated coefficients (exact sample spectrum) and PC-permutation (exact per-component marginals). The Gaussian and PC-permutation constructions agree; the exact-sample-spectrum construction yields smaller excesses for low-effective-rank clouds on the small- C sets (most visibly DINOv2 on CIFAR-100), so the size of those excesses is partly a property of the null hypothesis, while signs and the ImageNet readings are stable.

A.14 ORC EDGE TYPES

model	dataset	ORC within	ORC across	%neg within	%neg across
ViT-T	IN	0.35	0.27	5.2	9.4
ViT-T	C100	0.51	0.42	1.4	2.7
ViT-S	IN	0.37	0.28	3.4	8.3
ViT-S	C100	0.61	0.48	0.0	3.0
ViT-B	IN	0.40	0.30	2.1	5.9
ViT-B	C100	0.61	0.48	0.0	1.7
ViT-L	IN	0.40	0.31	2.1	6.8
ViT-L	C100	0.61	0.47	0.6	2.1
DINO-B	IN	0.40	0.33	2.8	4.1
DINO-B	C100	0.60	0.45	0.0	1.4
DINOv2-S	IN	0.37	0.33	1.8	1.2
DINOv2-S	C100	0.60	0.44	0.0	0.8
DINOv2-B	IN	0.58	0.64	0.3	0.1
DINOv2-B	C100	0.66	0.54	0.0	0.1
DINOv2-L	IN	0.69	0.72	0.0	0.0
DINOv2-L	C100	0.70	0.64	0.0	0.1
DINOv2-G	IN	0.70	0.74	0.0	0.0
DINOv2-G	C100	0.77	0.70	0.0	0.0
CLIP-B	IN	0.36	0.29	4.2	7.9
CLIP-B	C100	0.60	0.46	0.0	3.1
CLIP-L	IN	0.35	0.27	4.0	8.4
CLIP-L	C100	0.57	0.43	0.0	2.2
SigLIP-B	IN	0.34	0.26	5.0	7.5
SigLIP-B	C100	0.59	0.43	0.0	2.3

Table 33: Edge-type ORC on the centroid kNN graph (30 WordNet superclasses): negative curvature concentrates on between-cluster bridges.

A.15 PROMPT ROBUSTNESS

model	photo	name	image	closeup	this is	wild	def.	gloss only	concept	discuss.
BGE-base	0.124	0.118	0.125	0.126	0.125	0.115	0.119	0.115	0.119	0.121
E5-base	0.114	0.119	0.117	0.120	0.115	0.116	0.108	0.115	0.120	0.116
GPT-2	0.131	0.145	0.110	0.086	0.080	0.089	0.105	0.124	0.112	0.105
GPT-2-M	0.091	0.122	0.110	0.100	0.134	0.104	0.081	0.107	0.112	0.098

Table 34: $\hat{\delta}$ across prompt templates (ImageNet class names; “gloss only” contains no class name).

template	GPT-2 S	GPT-2 M	GPT-2 L	GPT-2 XL
photo	−0.016*	−0.019*	−0.043*	−0.050*
name_only	−0.006	−0.007	−0.048*	−0.051*
image	−0.018*	−0.028*	−0.050*	−0.049*
closeup	+0.007	−0.018*	−0.044*	−0.049*
this_is	−0.032*	−0.022*	−0.042*	−0.046*
wild	−0.023*	−0.067*	−0.014	−0.010*
def_pair	−0.002	−0.019*	−0.036*	−0.035*
gloss_only	−0.020*	−0.030*	−0.033*	−0.036*
concept	−0.002	−0.003	−0.043*	−0.041*
discussion	−0.010	−0.013*	−0.041*	−0.040*

Table 35: Template-conditioned excess over the spectrum-matched null for the four GPT-2 sizes (10 templates; *: $|z| \geq 2$ against combined null and estimator noise). The scale trend of §4 is assessed on every template, not only on the paper’s baseline. Padding-free extraction (one prompt at a time, the canonical protocol of Table 12). Mean excess over templates: −0.012 (S), −0.023 (M), −0.040 (L), −0.041 (XL); cells at $|z| \geq 2$: 5, 8, 9 and 10 of 10. Only 3 null replicates per cell here, so the z are coarser than the census’s 20-replicate values: on the baseline template the 200-replicate census (Table 12) finds S and M not genuine (excess −0.013/−0.009, $p = 0.09/0.14$), so their starred cells are borderline, whereas the L/XL cells (z from −4 to −9) are not in doubt. Template sensitivity of raw $\hat{\delta}$: range up to 0.065 for causal LMs and ≤ 0.012 for embedders (prompt table in Appendix 34).

A.16 SAMPLE-LEVEL FEATURES

model	CIFAR-100 (10/class)			DTD (22/class)		
	raw $\hat{\delta}$	excess	z	raw $\hat{\delta}$	excess	z
ViT-T	0.130	−0.0151	−1.7	0.124	−0.0127	−2.1
ViT-S	0.121	−0.0069	−0.8	0.103	−0.0165	−2.4
ViT-B	0.121	−0.0075	−0.9	0.092	−0.0190	−3.0
ViT-L	0.100	−0.0148	−2.9	0.092	−0.0108	−1.2
DINO-B	0.104	−0.0106	−1.7	0.110	−0.0069	−1.5
DINOv2-S	0.100	−0.0091	−1.3	0.145	+0.0126	+1.7
DINOv2-B	0.089	−0.0063	−1.1	0.139	+0.0233	+2.1
DINOv2-L	0.087	−0.0068	−1.5	0.131	+0.0201	+1.5
DINOv2-G	0.102	+0.0048	+0.4	0.136	+0.0232	+2.3
CLIP-B	0.117	−0.0176	−2.3	0.125	−0.0085	−1.2
CLIP-L	0.132	+0.0022	+0.2	0.120	−0.0108	−1.3
SigLIP-B	0.113	−0.0194	−2.3	0.114	−0.0152	−1.7

Table 36: The instrument transplanted to sample-level features, the object of prior latent-hyperbolicity readings (≈ 1000 stratified training images per cell, spectrum-matched nulls). Raw sample-level $\hat{\delta}$ sits in the same low band those works report, but the excess is mostly within null noise (6/24 cells at $|z| \geq 2$; DINOv2 on DTD is sign-positive): raw sample-level readings are confounded exactly as raw centroid readings are, and the genuine tree excess of the census is a statement about *inter-class* geometry.

A.17 TEXT-MODALITY TRANSFER

model	$\hat{\delta}$	excess	trip. (cos)	NC H-R (pp)	FS H-R (pp)	CI95
BGE-base	0.122	-0.028	0.89	-0.59	+0.05	± 0.05
E5-base	0.127	-0.024	0.89	-0.43	+0.04	± 0.03
GTE-base	0.132	-0.020	0.89	-0.82	+0.08	± 0.06

Table 37: DBpedia Classes (219 leaf classes, 3 levels): genuine tree excess and strong sibling alignment, yet $\hat{\delta}$ above the low- δ band correctly predicts marginal gains.

A.18 HIERARCAPS MULTI-LEVEL HIERARCHY

model	$\rho(\text{level, radius})$	% monotone	trip. L1	trip. L2	$\hat{\delta}$ leaves	excess
BGE-base	$+0.80 \pm 0.31$	49.6	0.88	0.97	0.130	+0.003
E5-base	$+0.87 \pm 0.19$	53.9	0.90	0.97	0.139	+0.010
GTE-base	$+0.47 \pm 0.55$	24.0	0.90	0.97	0.135	+0.006
CLIP-B	$+0.64 \pm 0.43$	27.2	0.86	0.95	0.123	-0.011
CLIP-L	$+0.51 \pm 0.49$	18.7	0.86	0.96	0.117	-0.010
SigLIP-B	$+0.32 \pm 0.55$	9.8	0.83	0.91	0.138	+0.014

Table 38: HierarCaps test set (1000 four-level chains): radial ordering under the paper’s projection (chance for strict monotonicity = $1/24 \approx 4.2\%$; shuffle controls ≈ 0), sibling triplet agreement (cosine; chance 0.5), and leaf $\hat{\delta}$ with spectrum-null excess.

A.19 PARALLELOGRAM DEFECT ξ : A NORM-STRUCTURE DESCRIPTOR

For a triangle (a, b, c) with m the approximate midpoint of (b, c) (the minimizer of $D(\cdot, b) + D(\cdot, c) + |D(\cdot, b) - D(\cdot, c)|$), the parallelogram defect is $\xi = (D(a, m)^2 + \frac{1}{4}D(b, c)^2 - \frac{1}{2}D(a, b)^2 - \frac{1}{2}D(a, c)^2) / (2D(a, m) \text{diam})$, averaged over 2×10^4 sampled triangles: 0 in Euclidean space, positive on spheres, negative in hyperbolic space and on trees.

space / model	ξ (mean)	frac. negative	null ξ	excess ($\pm$ CI95)	z
iid Gaussian ($d=768$)	+0.092	0%	—	—	—
S^d (geodesic)	+0.113	0%	—	—	—
S^d (chord)	+0.115	0%	—	—	—
$\mathbb{H}^2$ region ($R=6$)	-0.194	96%	—	—	—
balanced tree (depth 9)	-0.239	96%	—	—	—
ViT-T	+0.027	25%	$+0.027 \pm 0.002$	-0.001 ± 0.003	-0
ViT-S	+0.027	23%	$+0.042 \pm 0.002$	-0.015 ± 0.005	-7
ViT-B	+0.033	16%	$+0.047 \pm 0.002$	-0.014 ± 0.004	-6
ViT-L	+0.028	21%	$+0.046 \pm 0.002$	-0.018 ± 0.005	-7
DINO-B	-0.008	55%	$+0.052 \pm 0.003$	-0.059 ± 0.005	-21
DINOv2-S	-0.020	66%	$+0.061 \pm 0.004$	-0.081 ± 0.008	-20
DINOv2-B	-0.049	83%	$+0.075 \pm 0.003$	-0.123 ± 0.005	-45
DINOv2-L	-0.068	90%	$+0.080 \pm 0.002$	-0.149 ± 0.004	-66
DINOv2-G	-0.070	89%	$+0.082 \pm 0.002$	-0.152 ± 0.004	-69
CLIP-B	+0.019	32%	$+0.024 \pm 0.002$	-0.005 ± 0.003	-3
CLIP-L	+0.028	28%	$+0.032 \pm 0.002$	-0.004 ± 0.004	-2
SigLIP-B	+0.036	20%	$+0.031 \pm 0.002$	$+0.005 \pm 0.004$	+2

Table 39: Parallelogram curvature-sign estimator ξ on reference geometries (top) and ImageNet centroids (bottom), with the spectrum-matched null, excess and z -score for each model (20 null replicates; noise combines null and estimator s.d.): matched nulls have positive ξ , so raw readings understate curvature.

model	ξ_{geo}	frac. neg.	excess	z
ViT-T	+0.031	0.24	−0.0048	−2.7
ViT-S	+0.065	0.08	+0.0092	+6.2
ViT-B	+0.067	0.07	+0.0144	+10.5
ViT-L	+0.074	0.04	+0.0113	+7.9
DINO-B	+0.085	0.05	+0.0014	+1.0
DINOv2-S	+0.100	0.02	+0.0033	+2.3
DINOv2-B	+0.108	0.01	+0.0046	+4.9
DINOv2-L	+0.108	0.01	+0.0040	+3.9
DINOv2-G	+0.094	0.01	−0.0114	−12.5
CLIP-B	+0.025	0.28	−0.0129	−7.1
CLIP-L	+0.033	0.25	−0.0120	−4.8
SigLIP-B	+0.051	0.14	+0.0025	+1.2

Table 40: ξ on the angular geometry: centroids L2-normalized, distances the spherical geodesic, spectrum null built on the normalized cloud and re-normalized (20 replicates; ImageNet centroid store). Probes whether the deep negative ξ -excess of DINOv2 on ImageNet (Table 39) is a norm-structure effect. Outcome: it is for S/B/L, whose angular ξ -excess turns sphere-leaning positive, while DINOv2-G and CLIP-B/L remain negative on the sphere: the Euclidean ξ signal of DINOv2 is largely carried by norm structure.

reference (no curvature unless stated)	ξ	frac. negative
iid Gaussian	+0.0919	0.00
Gaussian, lognormal radii ($\sigma=0.3$)	−0.0621	0.99
Gaussian, lognormal radii ($\sigma=0.6$)	−0.0607	1.00
Gaussian, lognormal radii ($\sigma=1.0$)	−0.0496	0.99
core + shell mixture	−0.3217	0.79
uniform sphere	+0.1154	0.00
sphere with radial jitter	−0.0571	1.00

Table 41: ξ on references with heterogeneous norms but no curvature ($n=1000$, $d=768$, 5 seeds). Norm dispersion alone drives ξ negative (lognormal radii, radial jitter on a sphere, core+shell): the estimator reads norm structure, not curvature, and the paper reports it only as such.

B ILLUSTRATIVE VISUALIZATION

Figure 9 is a low-dimensional illustration of the hub-and-spoke organization; every quantitative comparison in the paper is made in the full feature space (§4).

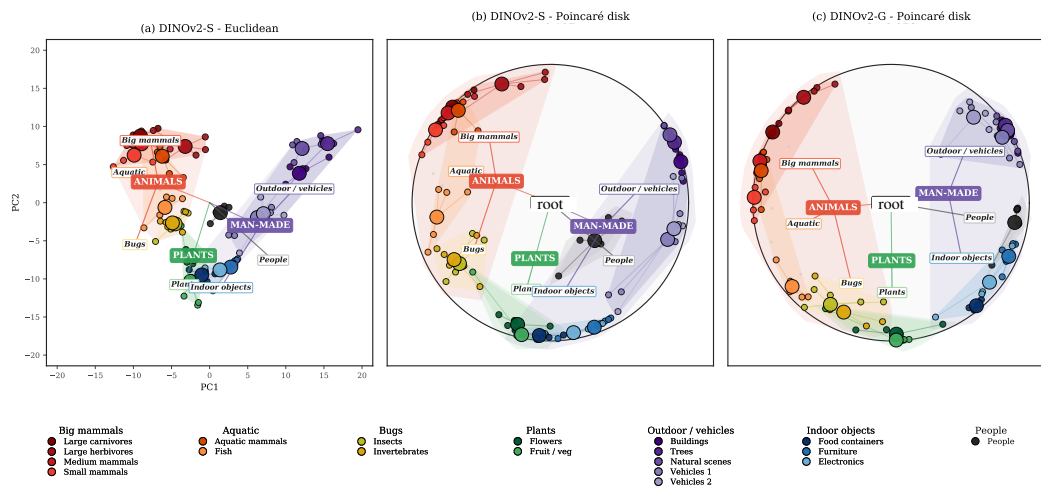


Figure 9: Illustrative 2-D projection of the 100 CIFAR-100 class centroids (per-panel PCA fit on the 20 super-class hubs; panels preserve 22–30% of hub variance and 6–10% of spoke length, so visual compactness is not comparable across panels and quantitative comparisons are made in full dimension). (a) DINOv2-S, Euclidean; (b) DINOv2-S after the projection of §3; (c) DINOv2-G.